



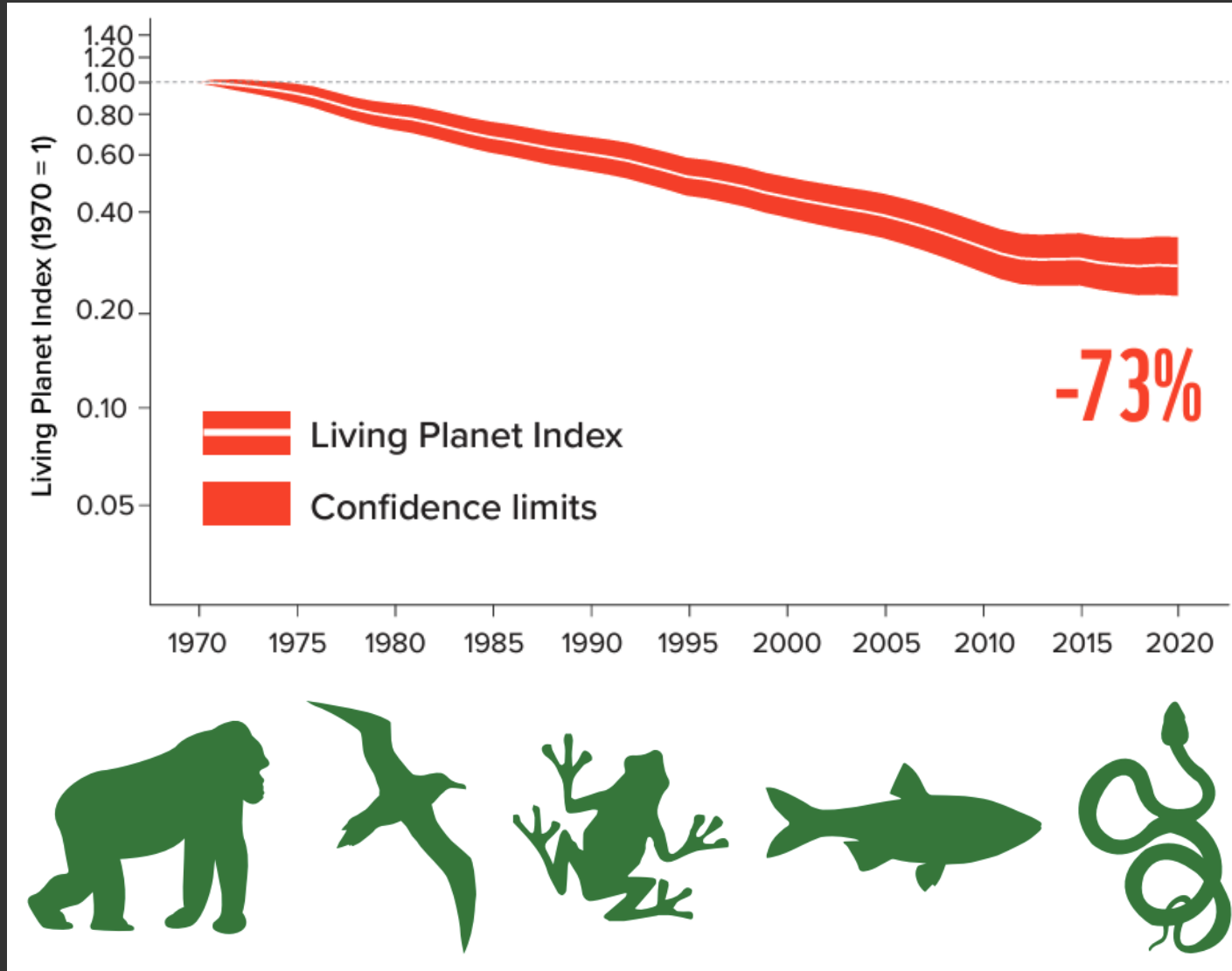
Accounting for adaptive variation when assessing biodiversity vulnerability to global environmental change

Only Razgour

O.Razgour@exeter.ac.uk

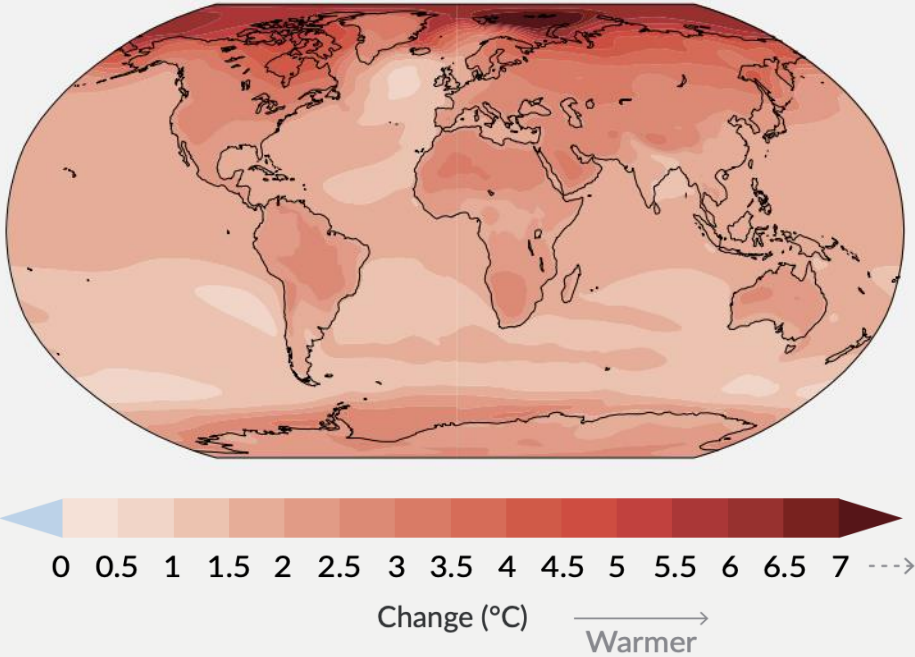
SSMPG, Sep. 2025

Global environmental change

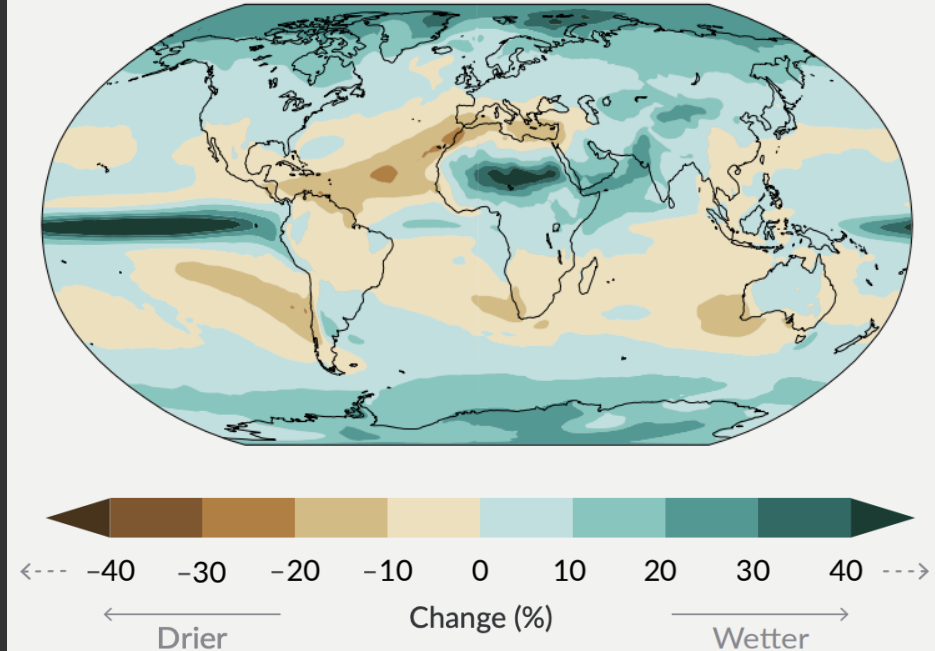


Global environmental change

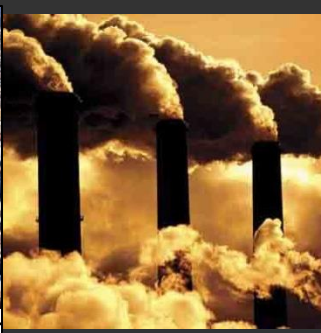
Simulated change at 2°C global warming



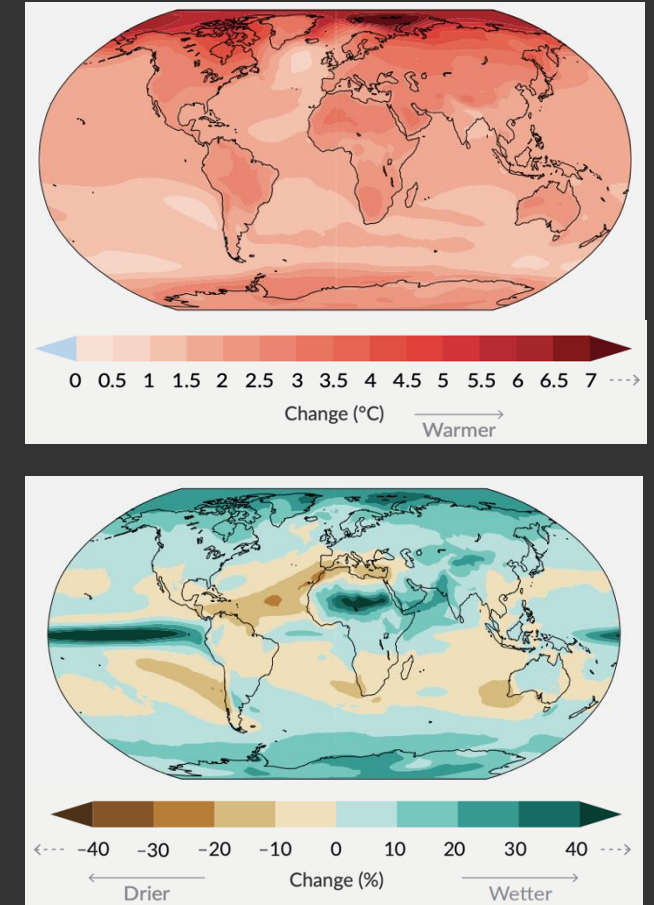
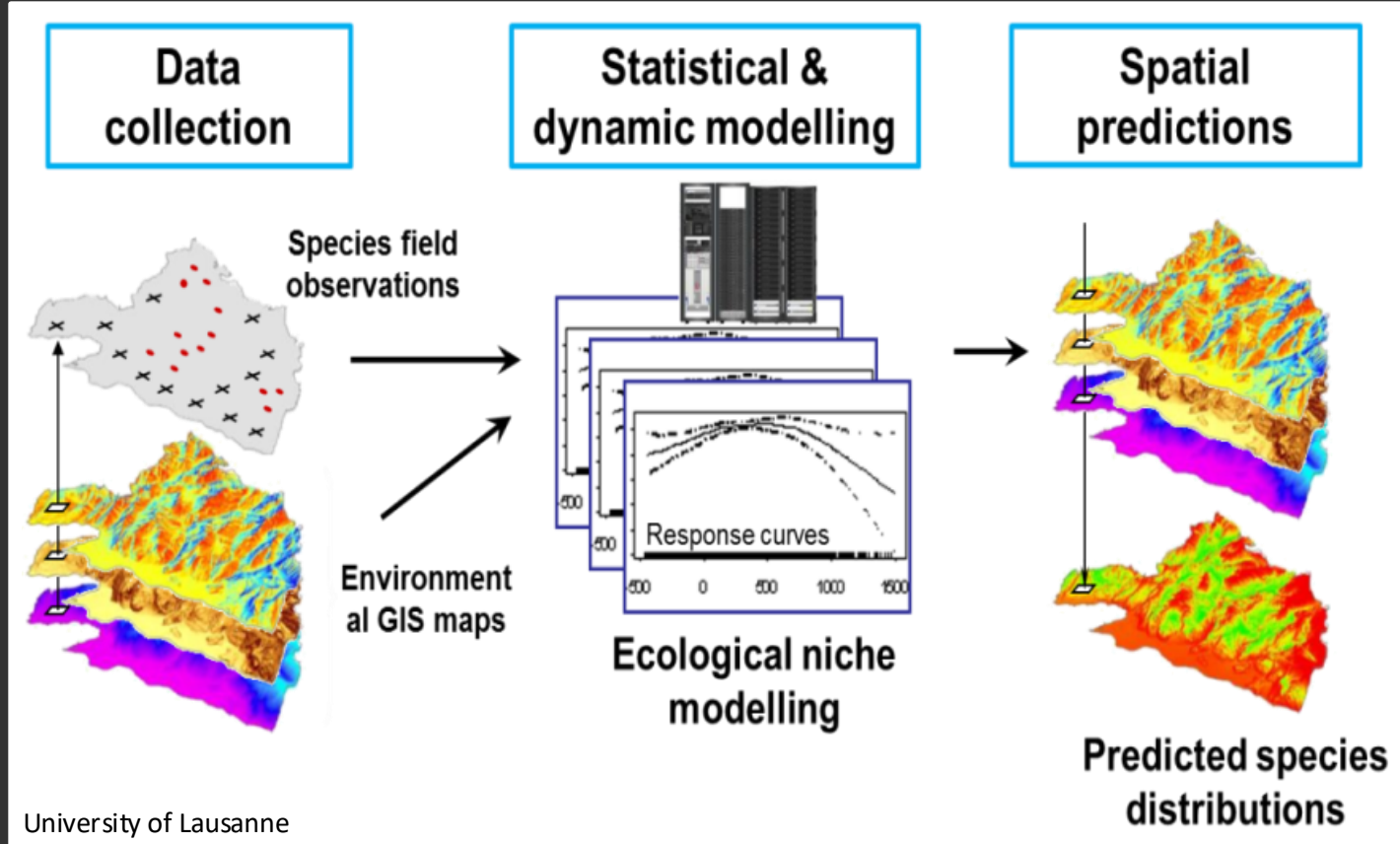
Simulated change at 2°C global warming



IPCC 6AR WGI 2021

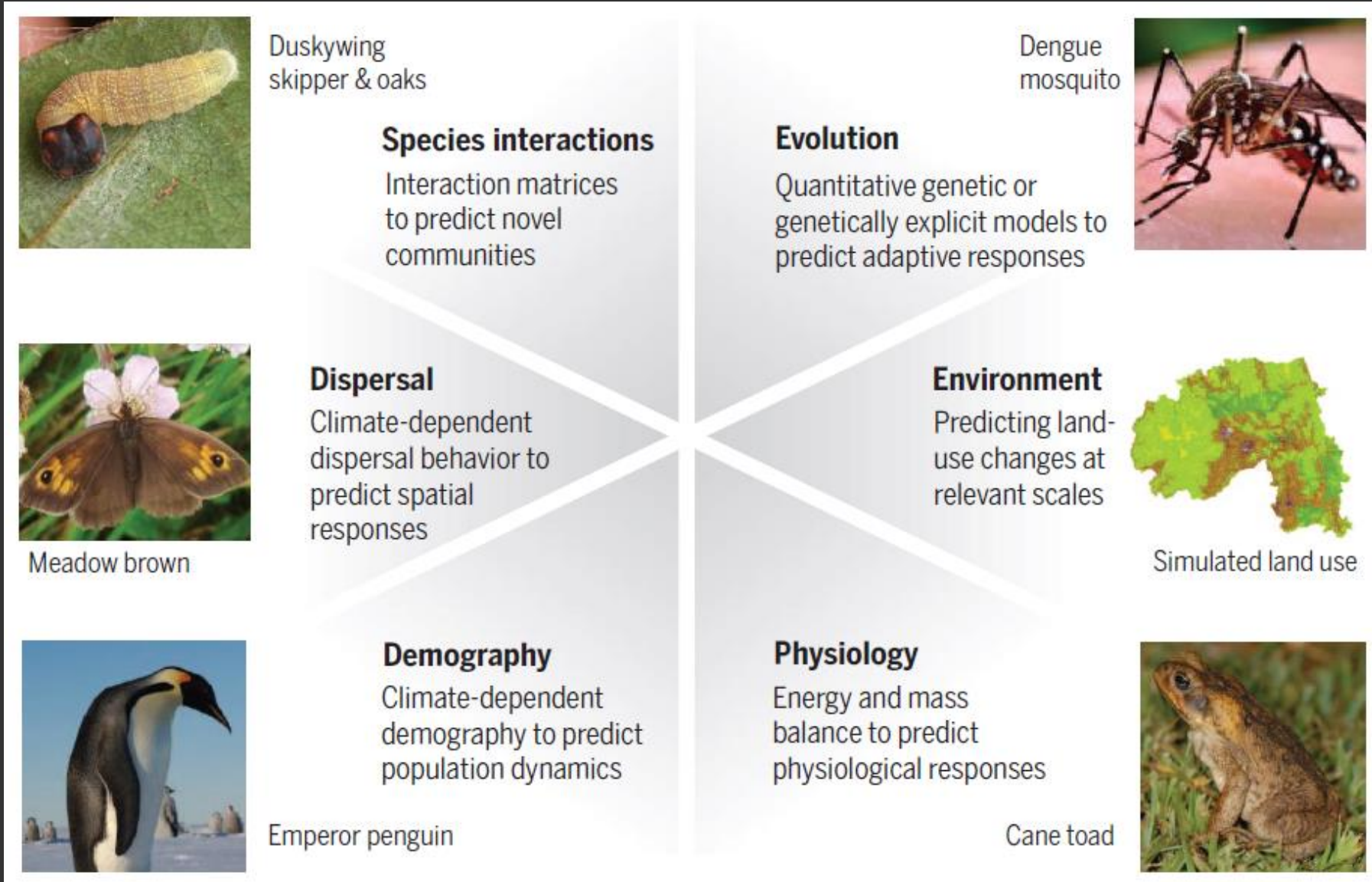


Biodiversity vulnerability assessment



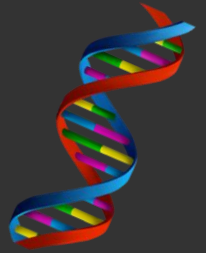
18-35% of species committed to extinction (Thomas et al. 2004, Nature)

Limitations of Species Distribution Models



(Urban et al.
2016, Science)

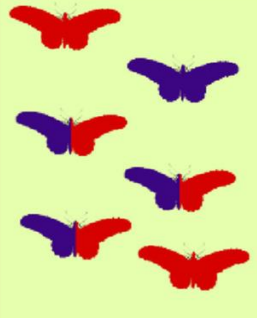
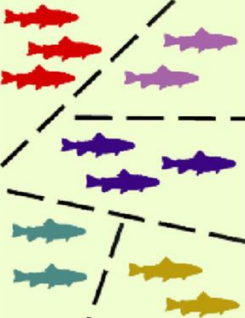
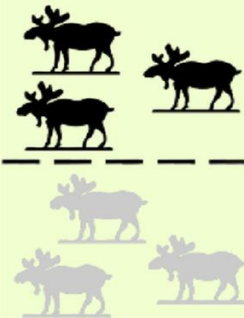
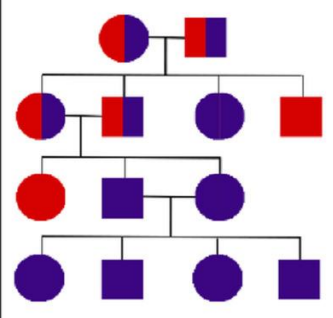
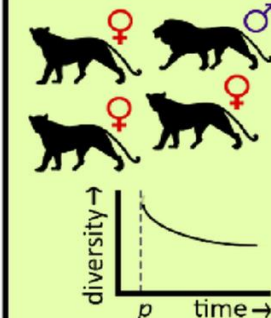
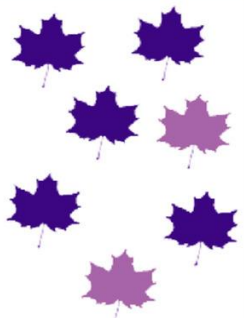
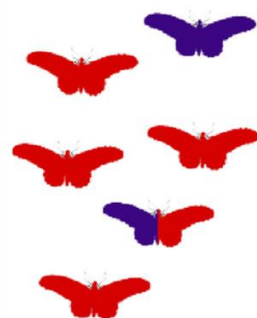
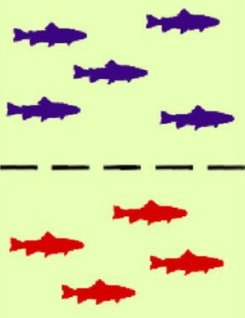
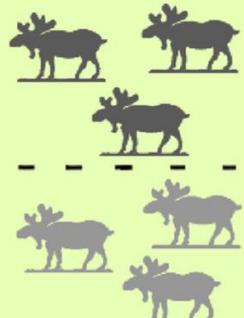
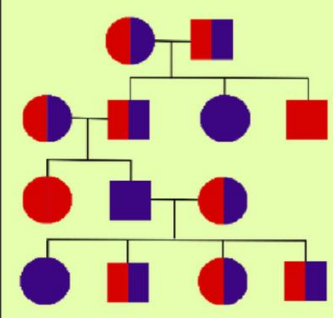
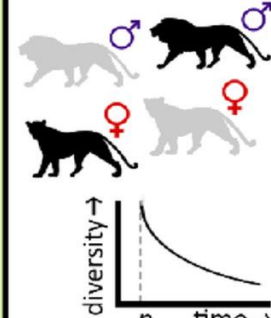
Genomic approaches



- Population conservation status
- Landscape and population connectivity
- Long-term viability
- Demographic history & baseline historic population sizes



Monitoring biodiversity responses to global change

Genetic Composition EBVs						
	Genetic diversity		Genetic differentiation		Inbreeding	Effective population size (N_e)
	Richness	Heterozygosity	Number of genetic units	Distance genetic units		
High						
Low						

Genomic offset = genomic vulnerability?



Southwestern willow flycatcher

High “genomic vulnerability”
(RCP 2.6, 2050)

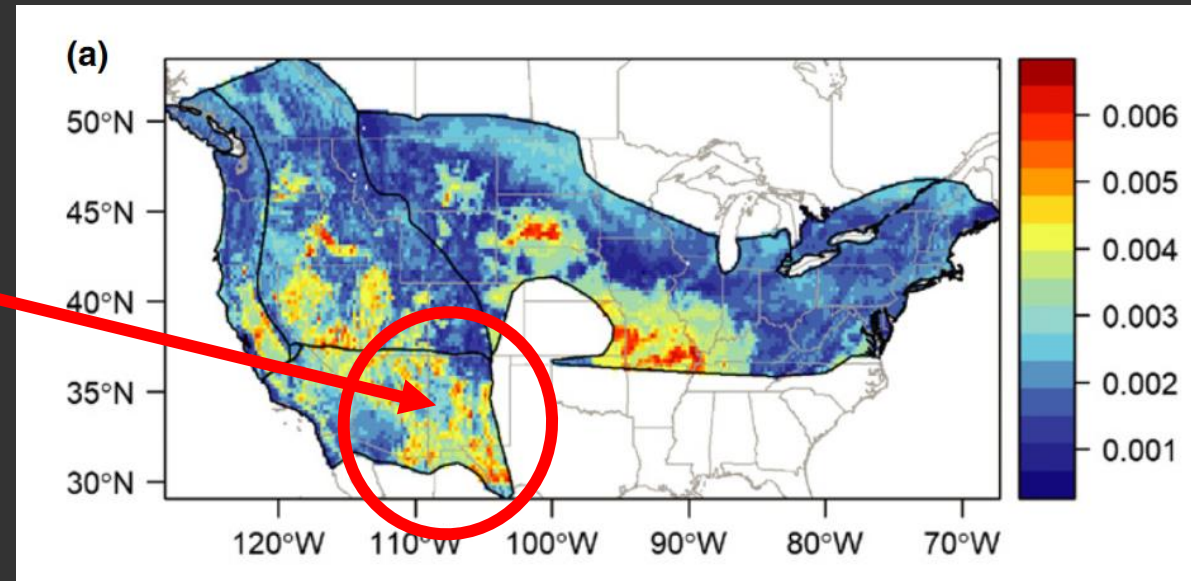
ECOLOGY LETTERS
Ecology Letters, (2018) 21: 1085–1096
doi: 10.1111/ele.12977

LETTER

Ecological genomics predicts climate vulnerability in an endangered southwestern songbird

Kristen Ruegg,^{1,2*} Rachael A. Bay,^{1,3,4†} Eric C. Anderson,^{3,4} James F. Saracco,⁵ Ryan J. Harrigan,¹ Mary Whitfield,⁶ Eben H. Paxton⁷ and Thomas B. Smith^{1,8}

Abstract
Few regions have been more severely impacted by climate change in the USA than the Desert Southwest. Here, we use ecological genomics to assess the potential for adaptation to rising global temperatures in a widespread songbird, the willow flycatcher (*Empidonax traillii*), and find the endangered desert southwestern subspecies (*E. t. extimus*) most vulnerable to future climate change. Highly significant correlations between present abundance and estimates of genomic vulnerability – the mismatch between current and predicted future genotype–environment relationships – indicate small, fragmented populations of the southwestern willow flycatcher will have to



Integrating dispersal into genomic offset analysis

Individual-based, spatially explicit eco-evolutionary simulation model

End of century, adaptive response to temperature change including dispersal

Journal of Heredity, 2023, XX, 1–13
<https://doi.org/10.1093/jhered/esac067>
Advance access publication 4 February 2023
Symposium article



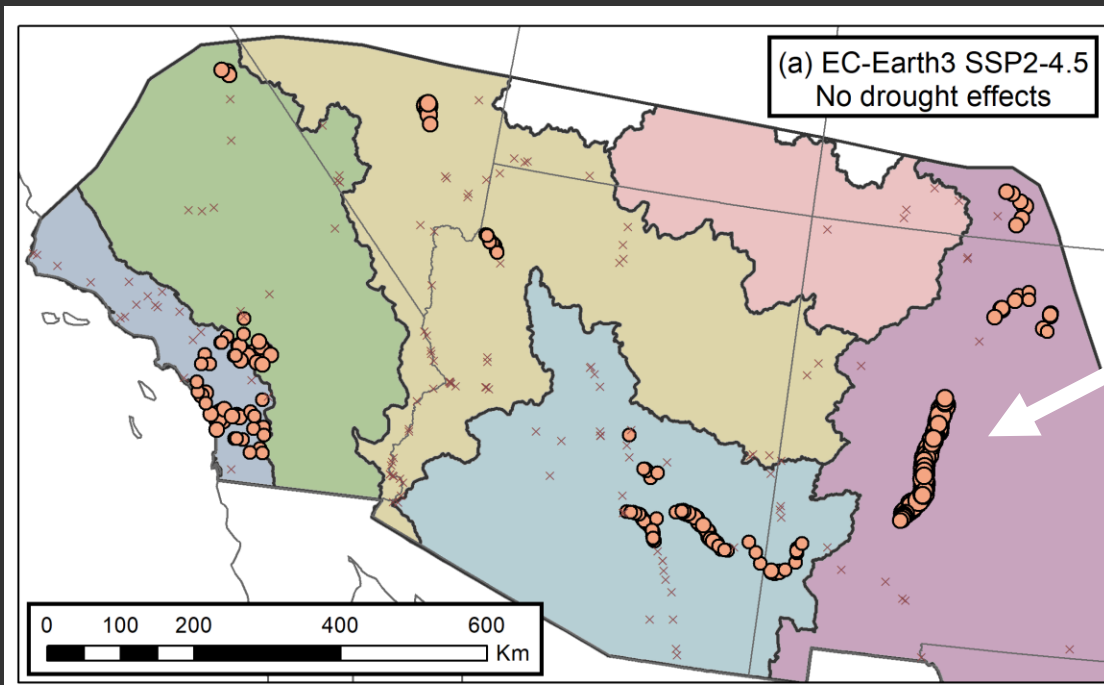
American
Genetic
Association

OXFORD

Symposium article

Evolutionary potential mitigates extinction risk under climate change in the endangered southwestern willow flycatcher

Brenna R. Forester^{1,4}, Casey C. Day², Kristen Ruegg¹, Erin L. Landguth^{2,3}



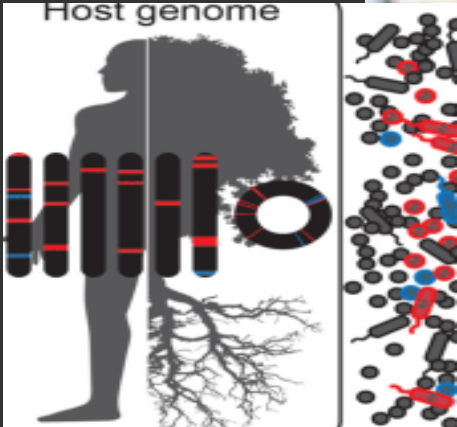
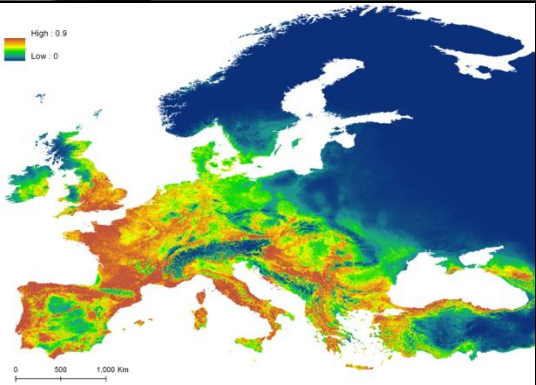
High “genomic vulnerability”

Patch Occupancy (2100)

- × Unoccupied
 - 1-5 birds
 - 6-10 birds
 - 11-15 birds
 - 16-20 birds
 - National & state boundaries
- | |
|--------------------|
| Coastal California |
| Basin and Mojave |
| Lower Colorado |
| Gila |
| Upper Colorado |
| Rio Grande |

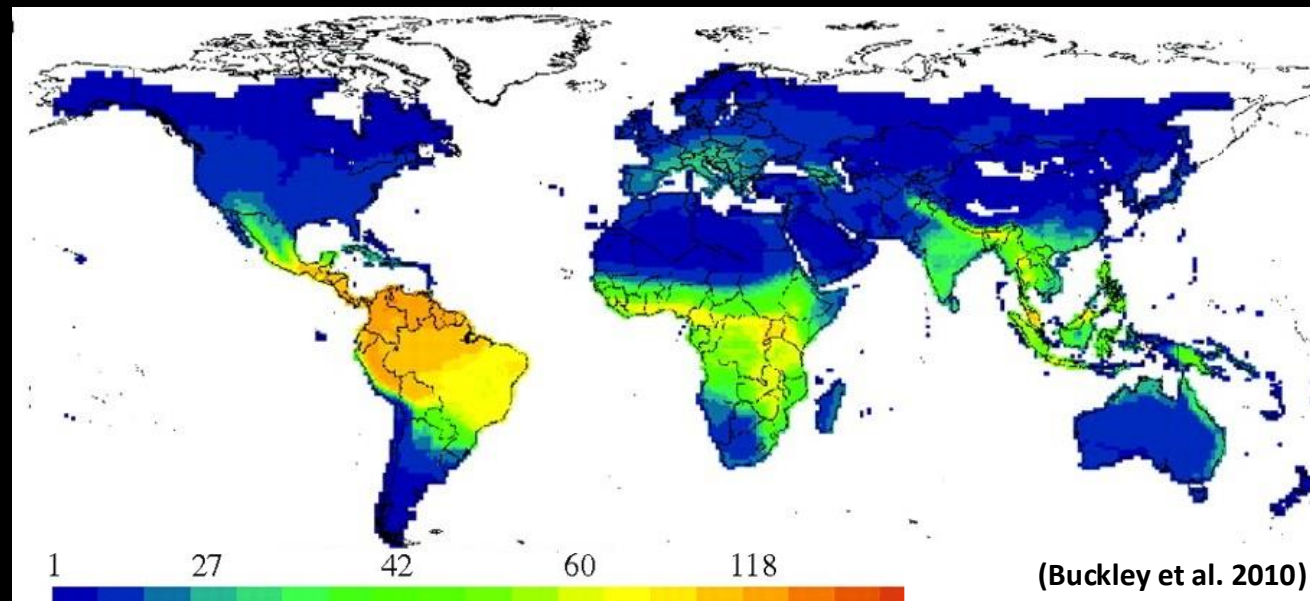


Integrated genomic approaches





IBR-1987330 - (c) - Ivan Kuzmin/imageBROKER



[Photos: Antton Alberdi; Daniel Hargreaves; Ivan Kuzmin; Paul Webala; Merlin Tuttle]

Talk Outline



1. Climate change vulnerability assessment
2. Responses to artificial lighting
3. Life on the Edge Toolbox
4. Q&A
5. Group exercise



Climate change vulnerability assessment

PNAS

Considering adaptive genetic variation in climate change vulnerability assessment reduces species range loss projections

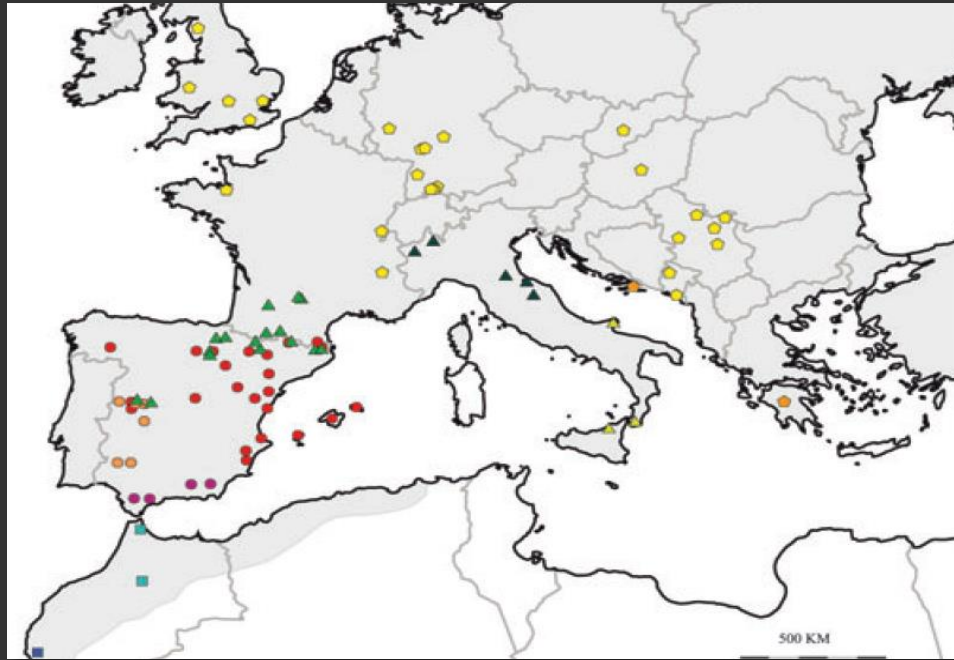
[Orly Razgour](#)  , [Brenna Forester](#), [John B. Taggart](#), [Michaël Bekaert](#) , [Javier Juste](#), [Carlos Ibáñez](#), [Sébastien J. Puechmaille](#), [Roberto Novella-Fernandez](#), [Antton Alberdi](#), and [Stéphanie Manel](#)  [-6](#) [Authors Info & Affiliations](#)



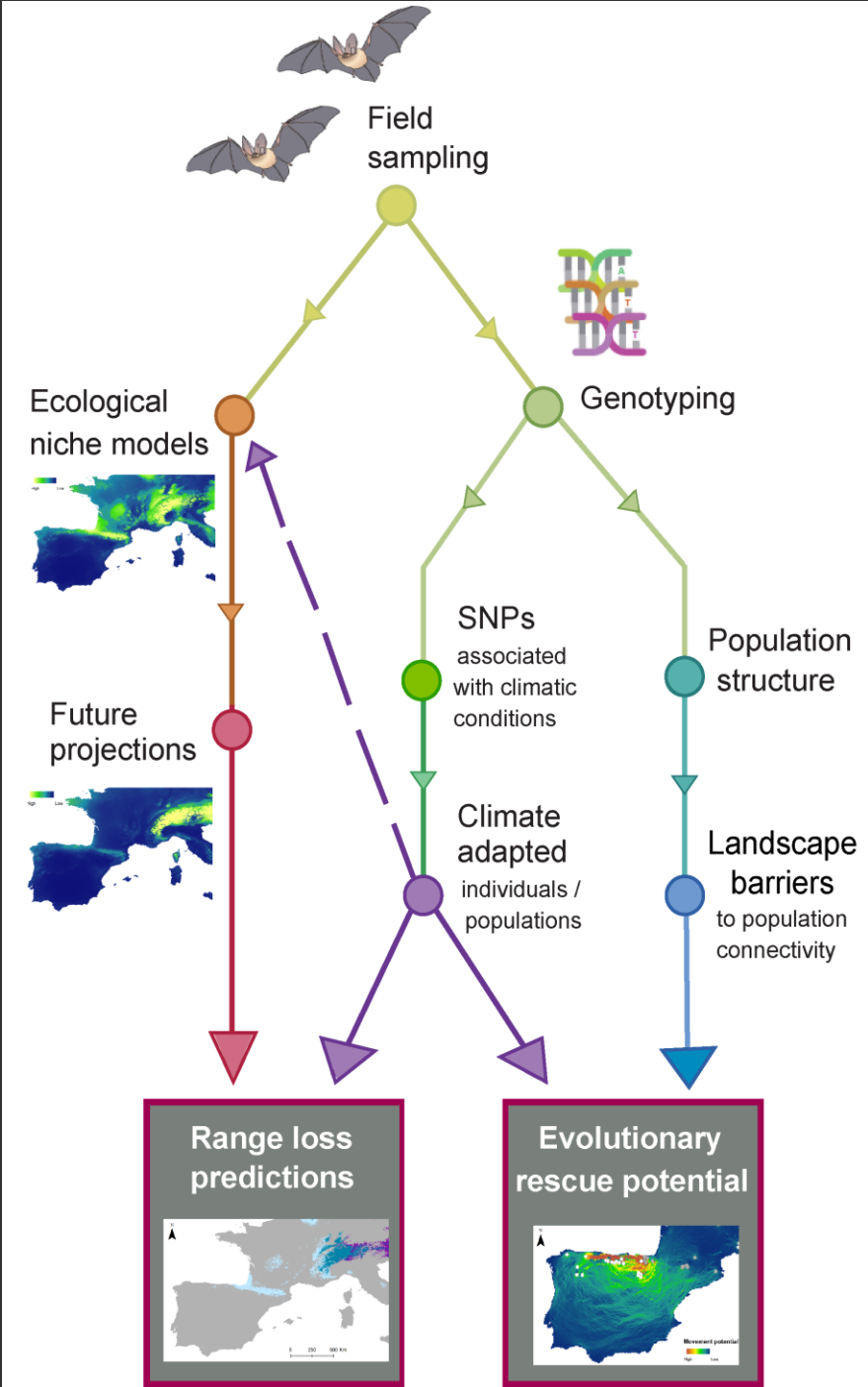
Brenna Forester



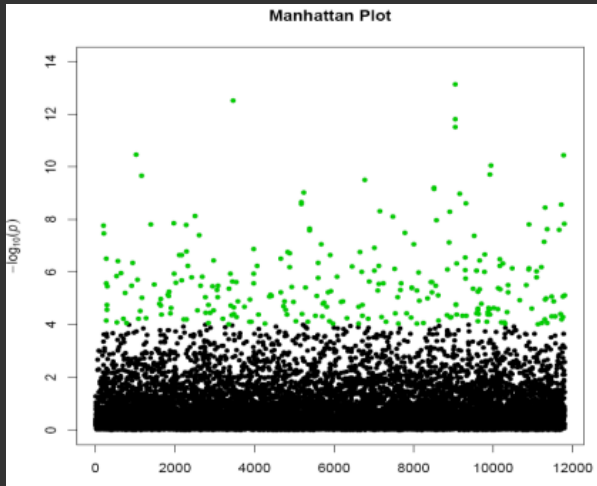
Incorporating adaptive variation into vulnerability assessment



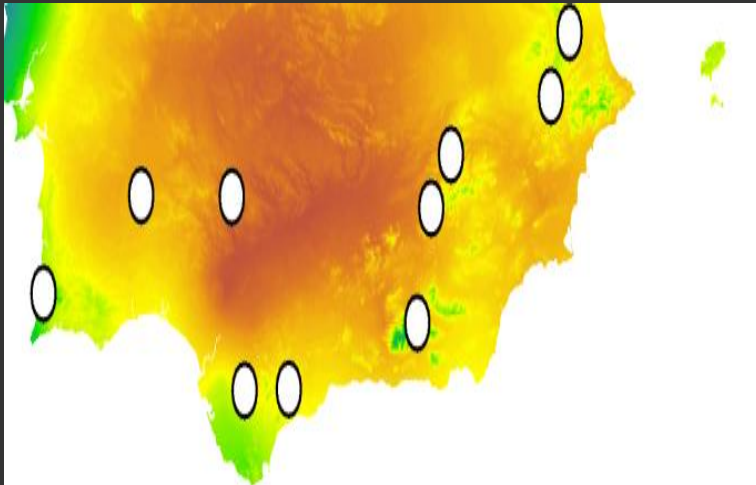
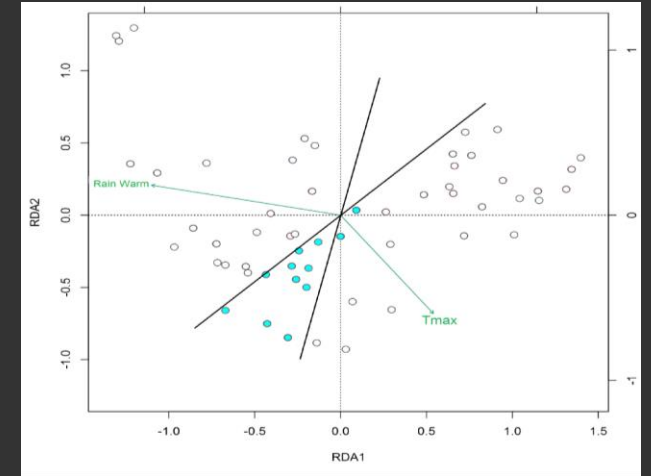
Genomic dataset: 20,000 SNPs



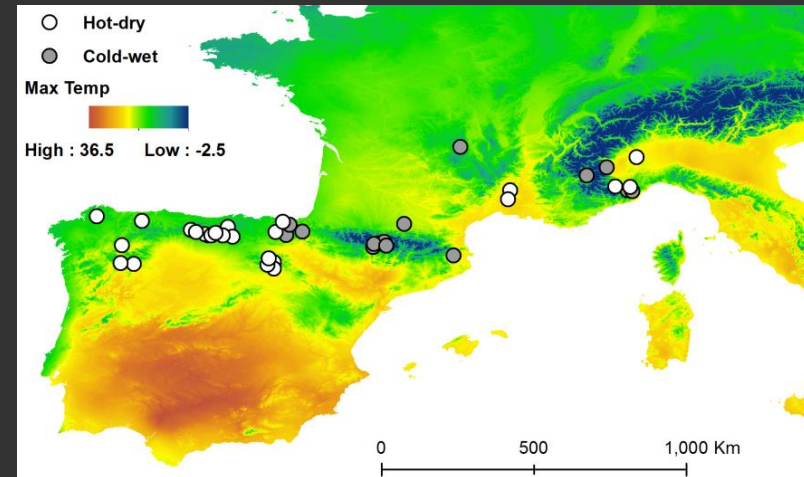
1. Identifying climate-adapted individuals



LFMM + RDA:
32 & 38 SNPs
associated with
max temperature
& summer rainfall



Myotis escaleraei



Myotis crypticus

2. Incorporating adaptations into SDMs

Environmental variables

BIO5 - max temperature of the warmest month

BIO6 - min temperature of the coldest month

BIO7 - temperature annual range

BIO12 - annual rainfall

BIO15 - rain seasonality

BIO16 - rainfall in the wet quarter

BIO18 - rainfall in the warmest quarter

Minimum April temperature

Slope

Karsts



Species distribution modeling,
calibration and evaluation,
ensemble modeling

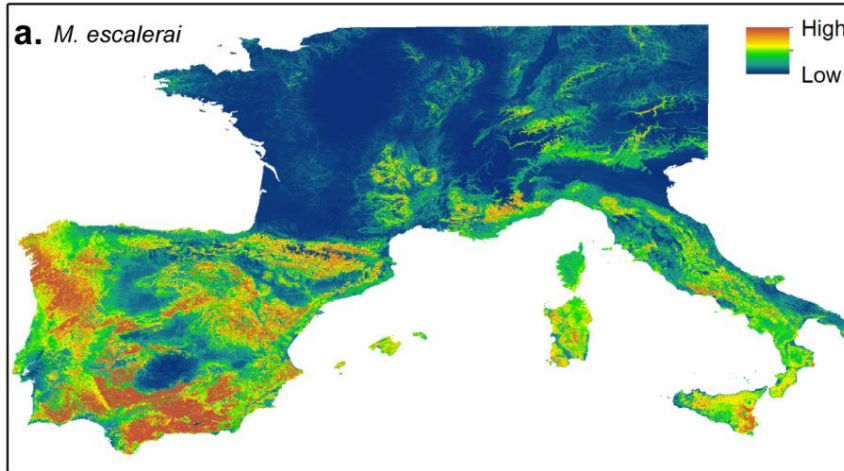
<https://biomodhub.github.io/biomod2/>

2. Incorporating adaptations into SDMs

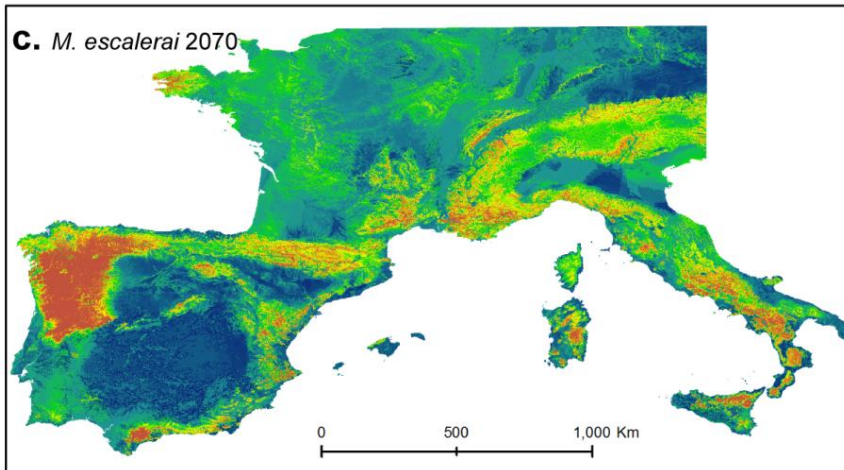
all records

hot-dry + cold-wet

Present



Future
(2070)



3. Effect of landscape barriers on evolutionary rescue potential



Landscape variables

Autumn temperature

Autumn rainfall

Forest cover

% tree cover

Land cover

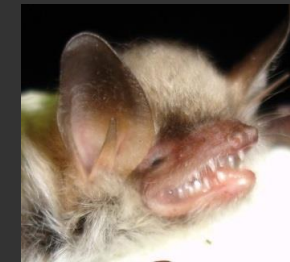
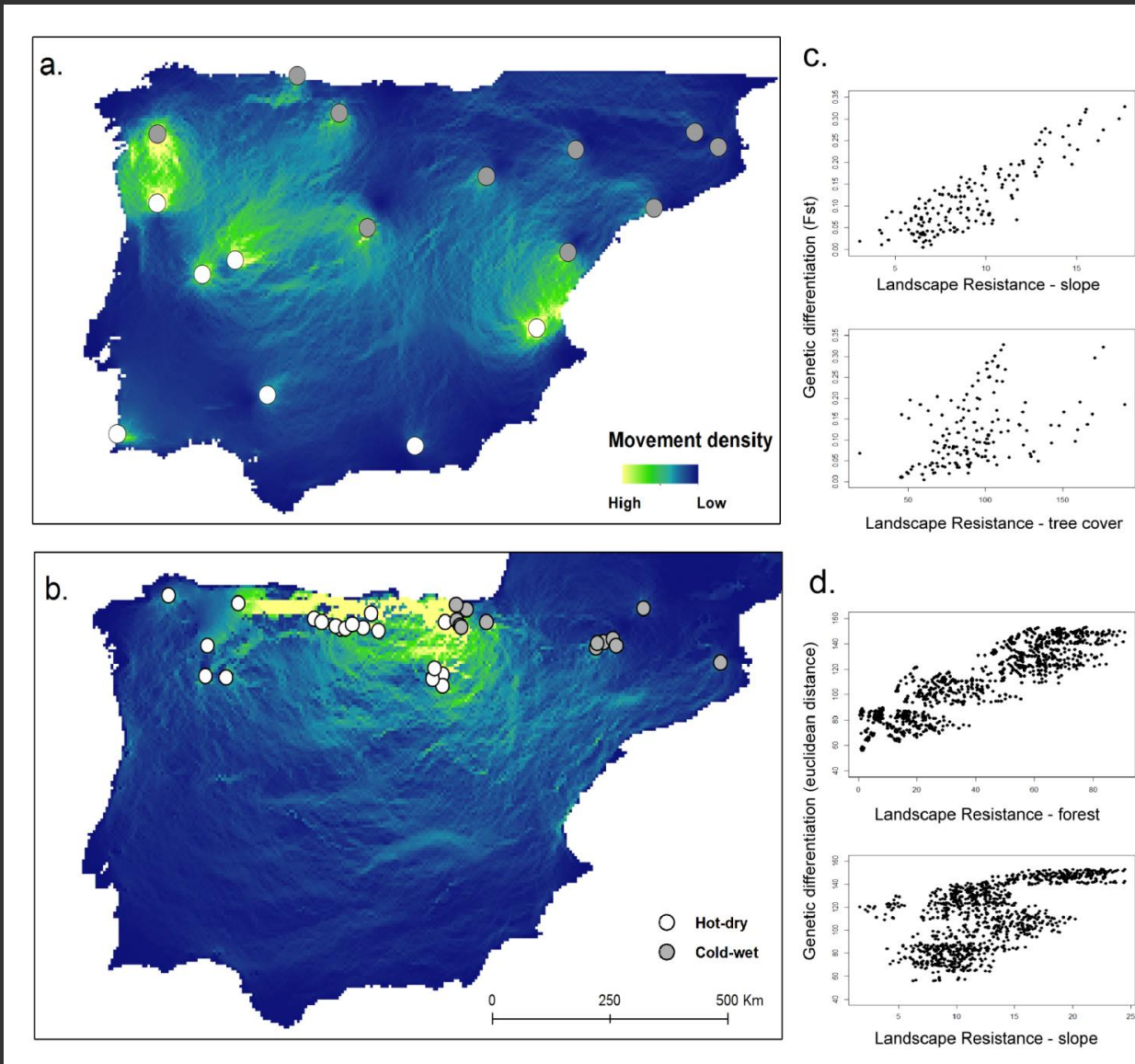
Habitat suitability

Altitude

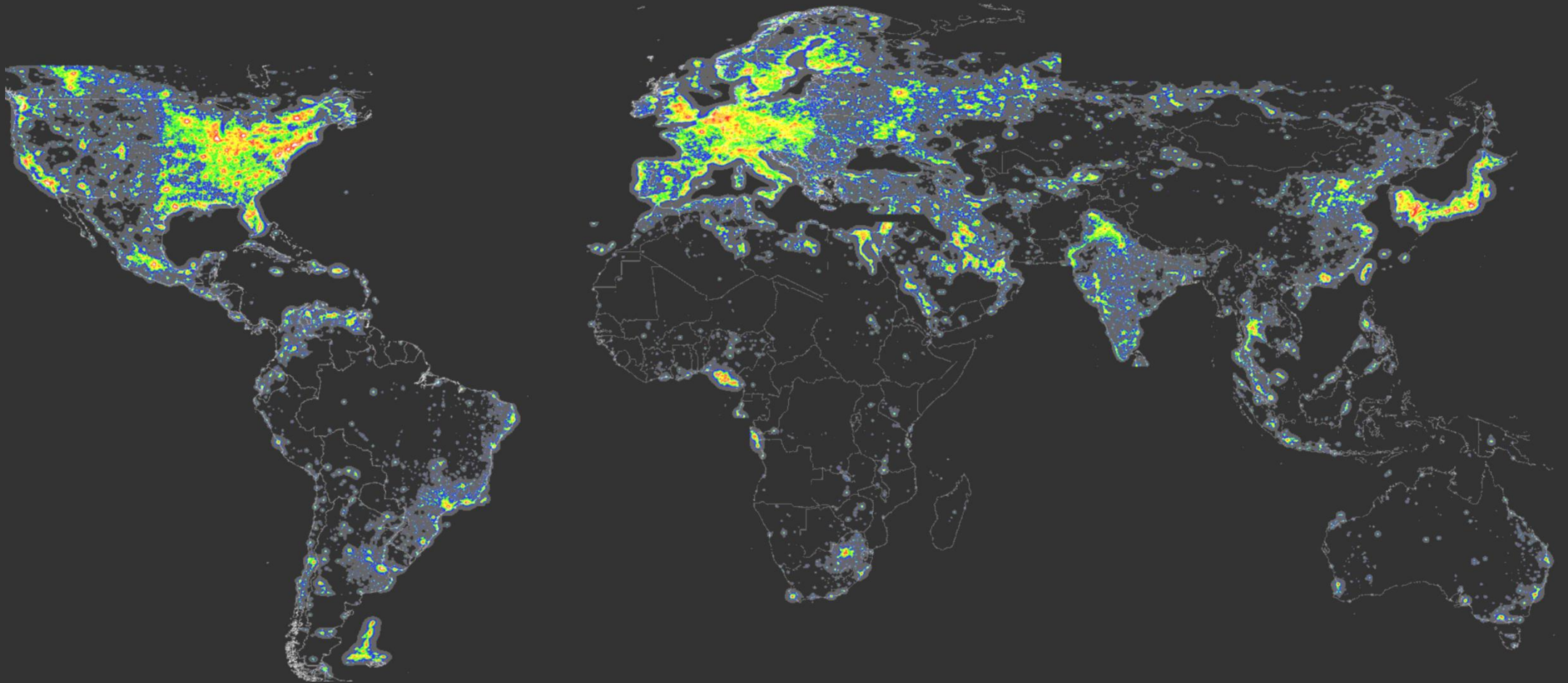
Slope

Topographic ruggedness

3. Effect of landscape barriers on evolutionary rescue potential



Multi-omic responses to artificial lighting



Understanding population level effects of artificial lighting on bats

- Ecological + genomic + physiological responses
- *Rhinolophus hipposideros* in Britain



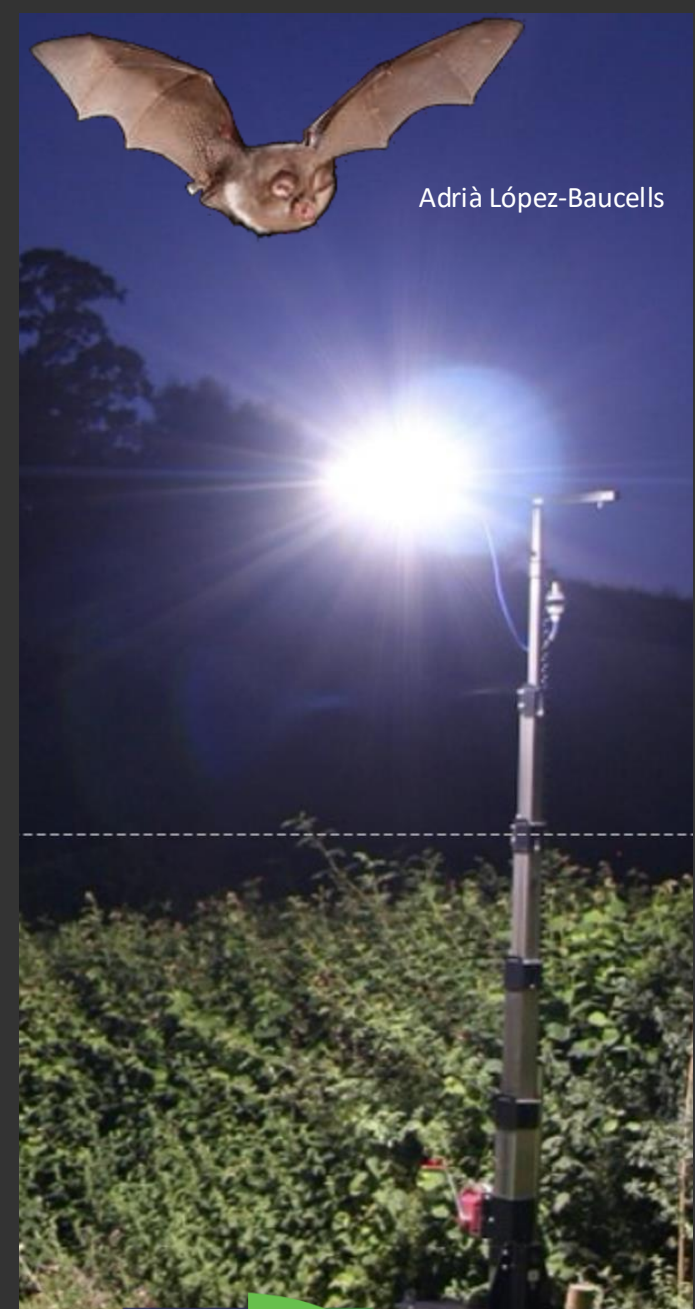
Emma Stone



Tom Foxley



Tom Jenkins

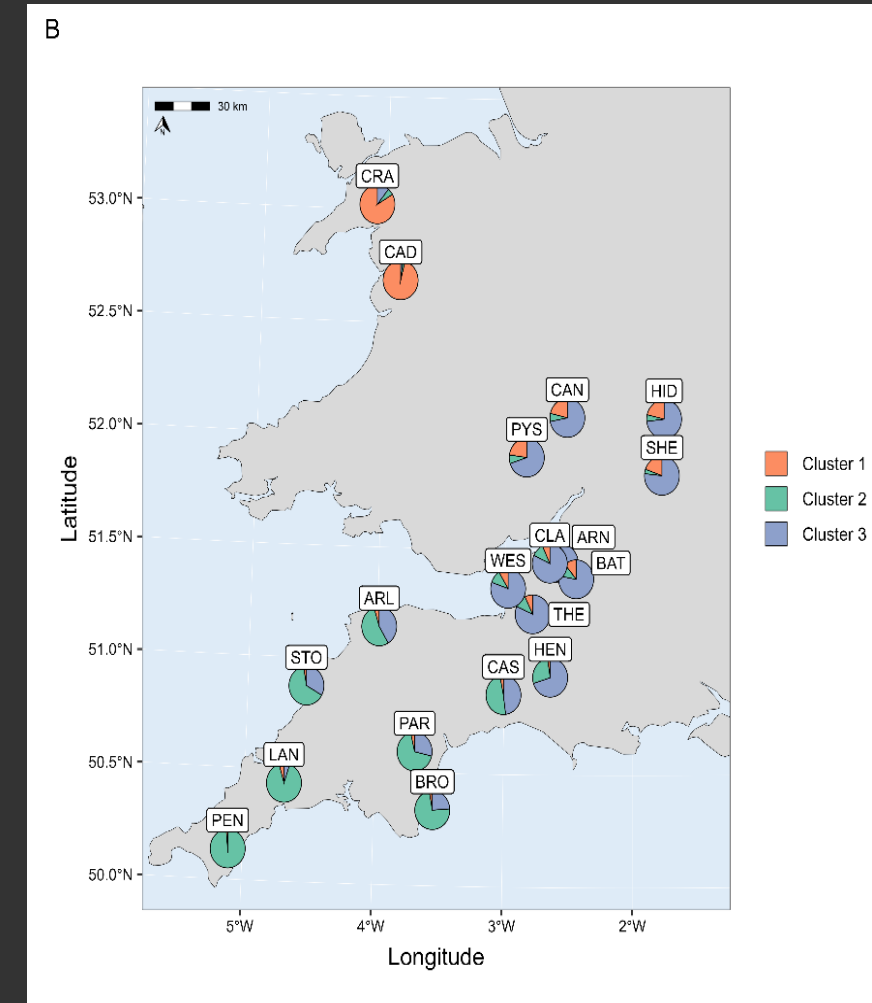
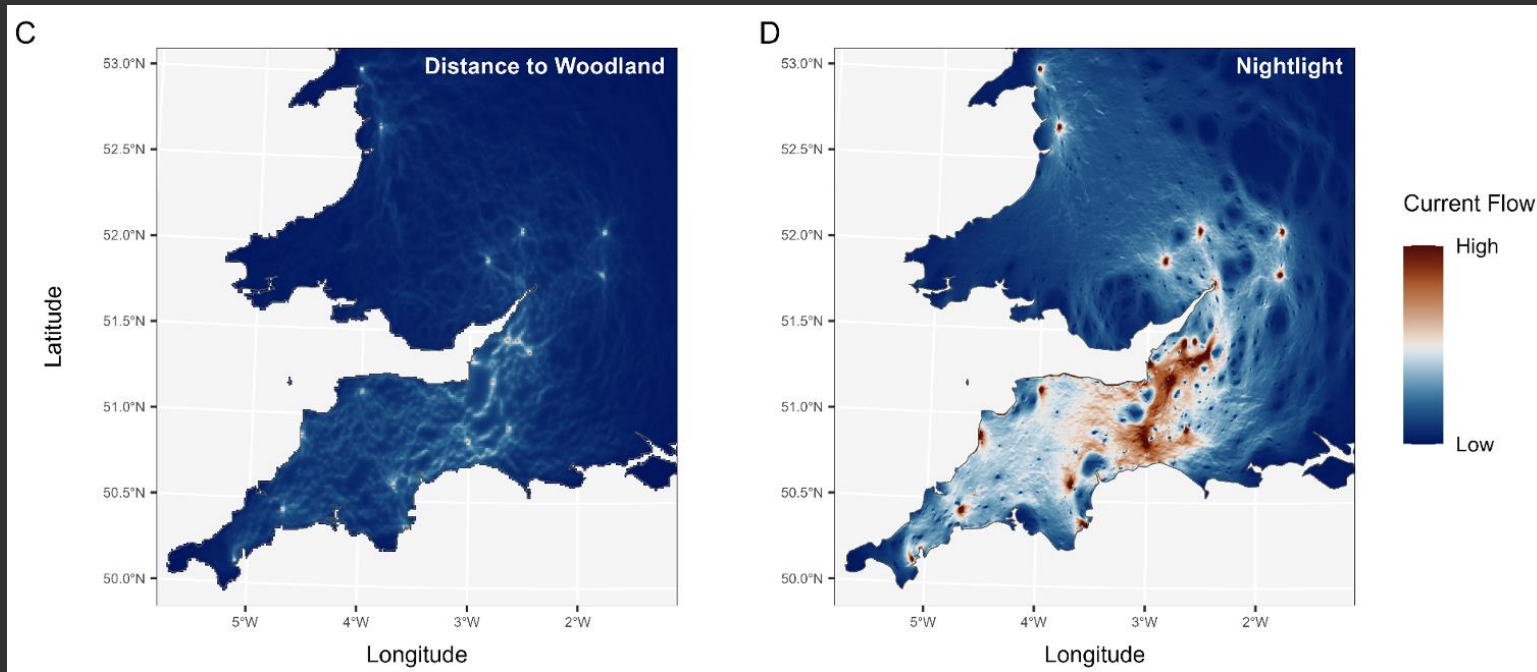


Adrià López-Baucells

Genetic composition under artificial lighting

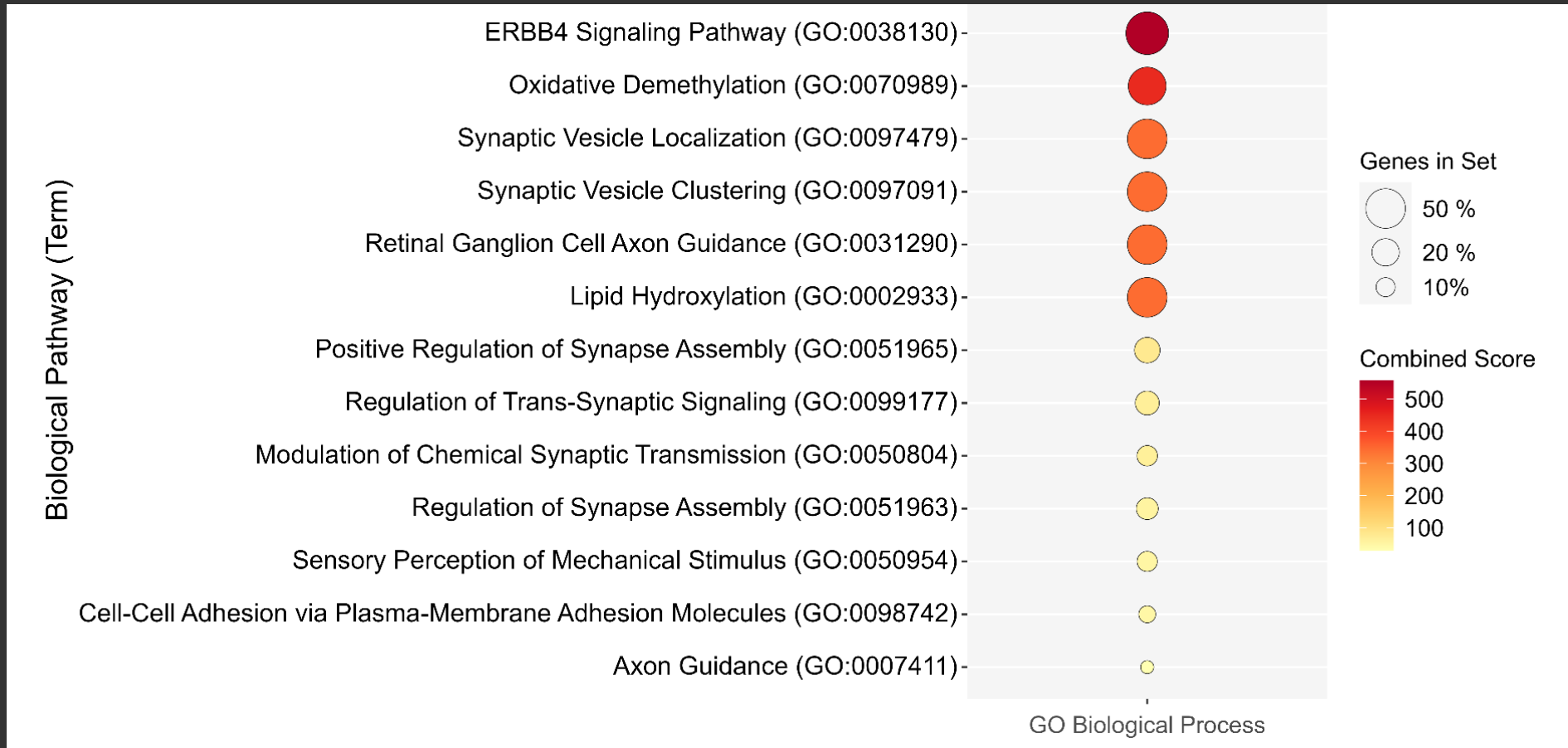


Whole genome sequencing 200 bats

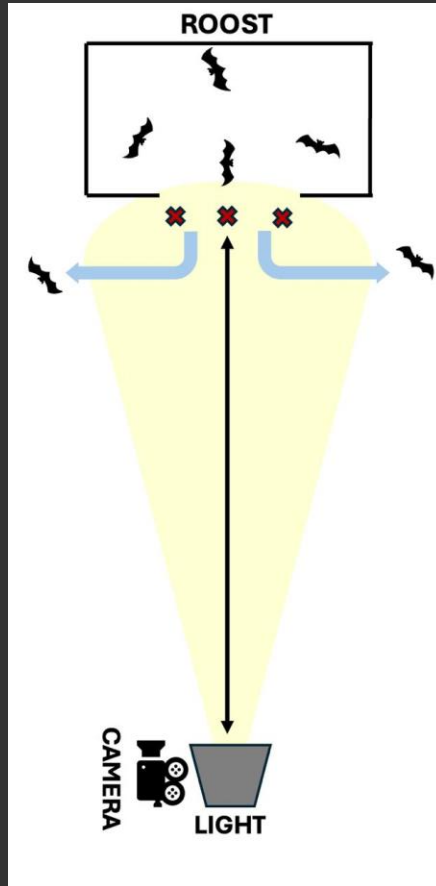


Adaptive responses to artificial lighting

Genes linked to visual and neural function



Behavioural responses to artificial lighting

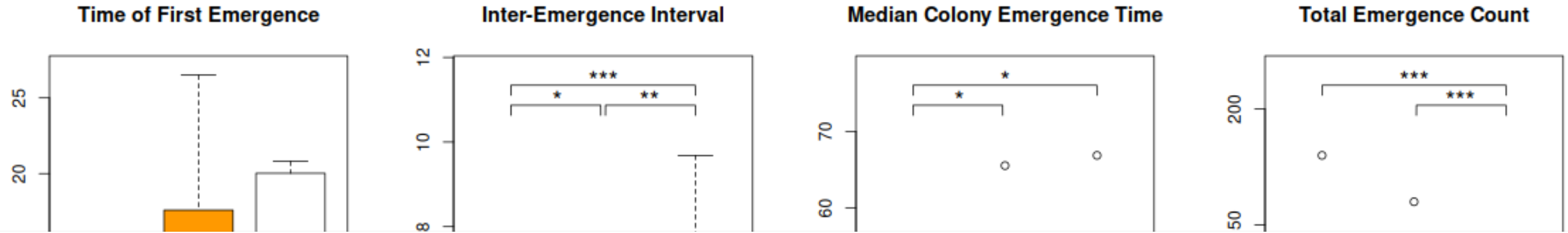


- Control – dark
- White LED light
- PC amber LED light

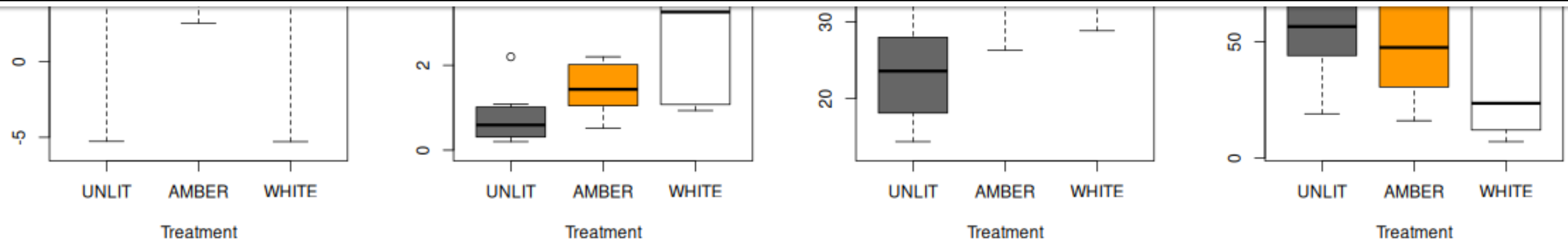


Jack Hooker

Behavioural responses to artificial lighting



Strong behavioural response to light exposure, especially white light



Genomic responses to artificial lighting

Whole transcriptome sequencing (RNA-seq)

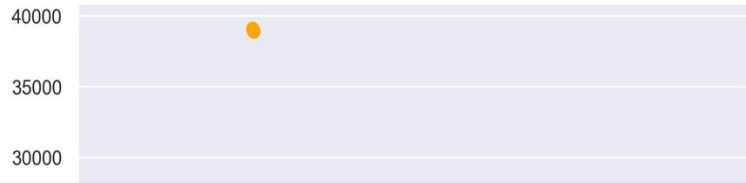
- Dark treatment: 10 bats
- Light treatment: 10 bats



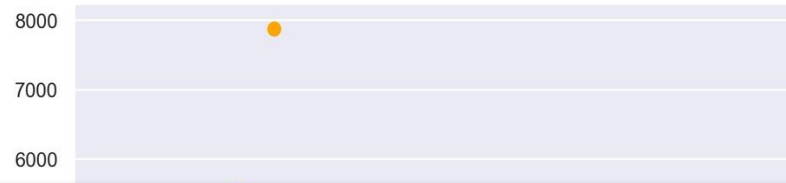
Transcriptomic responses to artificial lighting

252 Differentially Expressed Genes

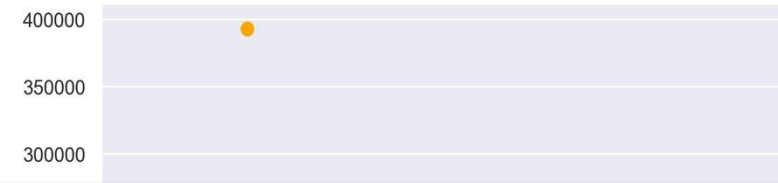
Gene expression: DDB1



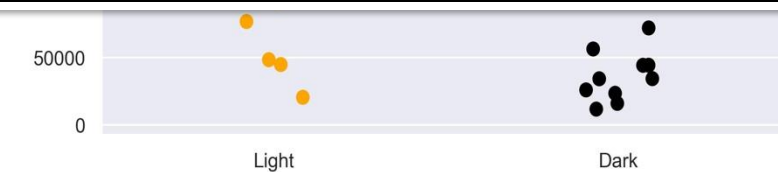
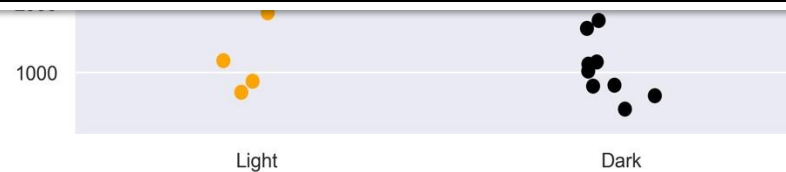
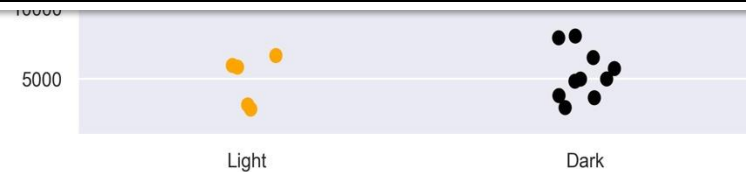
Gene expression: GADD45A



Gene expression: HBB



Some of the bats are mounting a strong response to light exposure



DDB1: UV damage recognition

GADD45A: DNA repair in response to UV-induced damage

HBB: Breakdown Hydrogen Peroxide – response to oxidative stress

(Jenkins et al., In Prep)



Chris Barratt



Methods in Ecology and Evolution

RESEARCH ARTICLE

Open Access



Life on the edge: A new toolbox for population-level climate change vulnerability assessments

Christopher D. Barratt , Renske E. Onstein, Malin L. Pinsky, Sebastian Steinfartz, Hjalmar S. Kühl, Brenna R. Forester, Orly Razgour



Volume 15, Issue 11

November 2024

Pages 2038-2058

https://cd-barratt.github.io/Life_on_the_edge.github.io/

Life On the Edge - Motivation

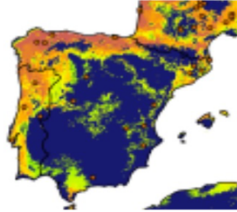
1. Build a generalized framework to predict population vulnerability to global change - applicable to any species or geographic area
2. Make this framework modular so that it can be easily expanded/edited as methods are updated/new ones are developed in future

LOtE: Structure of the framework

Identify populations under threat from global change

1

Exposure



Predictive distribution modelling

Species distribution modelling under current and future conditions to estimate geographic range changes

[Worldclim2/CHELSA](#), [Biomod2](#)

Environmental dissimilarity

Comparison between current and future conditions to quantify magnitude of predicted environmental change

2

Sensitivity



Adaptive sensitivity

Identify adaptive SNPs under selection and assess spatial distribution of local adaptations and genomic offset

[LFMM](#), [RDA](#), [genomic offset](#)

Neutral sensitivity

Estimate neutral (non-adaptive) genetic diversity [Plink](#)

3

Movement barriers

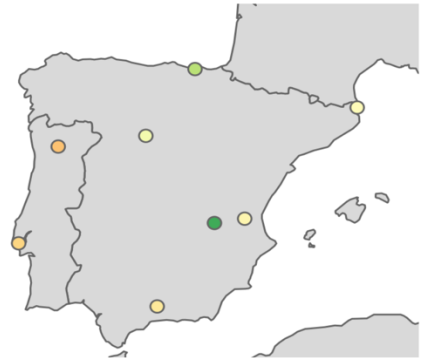


Predictive landscape genetics

Predict population connectivity based on parameterized resistance costs of landscape

[Circuitscape](#), [Julia](#)

Population vulnerability



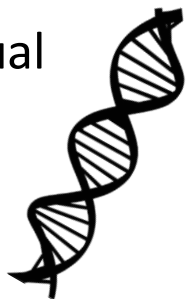
low
1
2
3
4
5
6
7
8
9
10
high

Vulnerability calculated for each population with data

Structure: inputs/processes/outputs

-data-

Population/individual
level genomic data
(Plink .ped & .map)



Geographic coordinates
(.csv format) and env.
predictor data (rasters)

Life on the edge

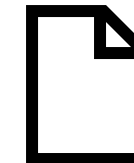
R, bash and
Julia functions



Params text file with ~60
parameters across analyses
(one line per species)

-outputs-

Structured and
standardised output
files and plots



Summary and log file

Structure: Running and output structure

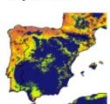
-analysis-



R, bash and Julia functions

```
prepare_spatial_data()
prepare_environmental_data()
spatially_rarefy_create_background_data()
sdms_biomod2()
exposure()
neutral_sensitivity()
impute_missing_data()
gea_lfmm()
gea_rda()
gea_rda_individual_categorization()
quantify_local_adaptation()
genomic_offset()
adaptive_sensitivity()
adaptive_sdms_biomod2()
create_circuitscape_inputs()
movement_barriers()
population_vulnerability()
summary_pdfs()
```

Exposure



Predictive distribution modelling
Species distribution modelling under current and future conditions to estimate geographic range changes
[Worldclim2/CHELSA](#), [Biomod2](#)

Environmental dissimilarity
Comparison between current and future conditions to quantify magnitude of predicted environmental change

Sensitivity



Adaptive sensitivity
Identify adaptive SNPs under selection and assess spatial distribution of local adaptations and genomic offset
[LFMM](#), [RDA](#), [genomic offset](#)

Neutral sensitivity
Estimate neutral (non-adaptive) genetic diversity [Plink](#)

Movement barriers



Predictive landscape genetics
Predict population connectivity based on parameterized resistance costs of landscape
[Circuitscape](#), [Julia](#)



Run_life_on_the_edge.sh

-outputs-

Exposure

SDMs

env_dissimilarity

Sensitivity

Neutral sensitivity

Adaptive_sensitivity

Movement_barriers

Circuitscape

Population_vulnerability



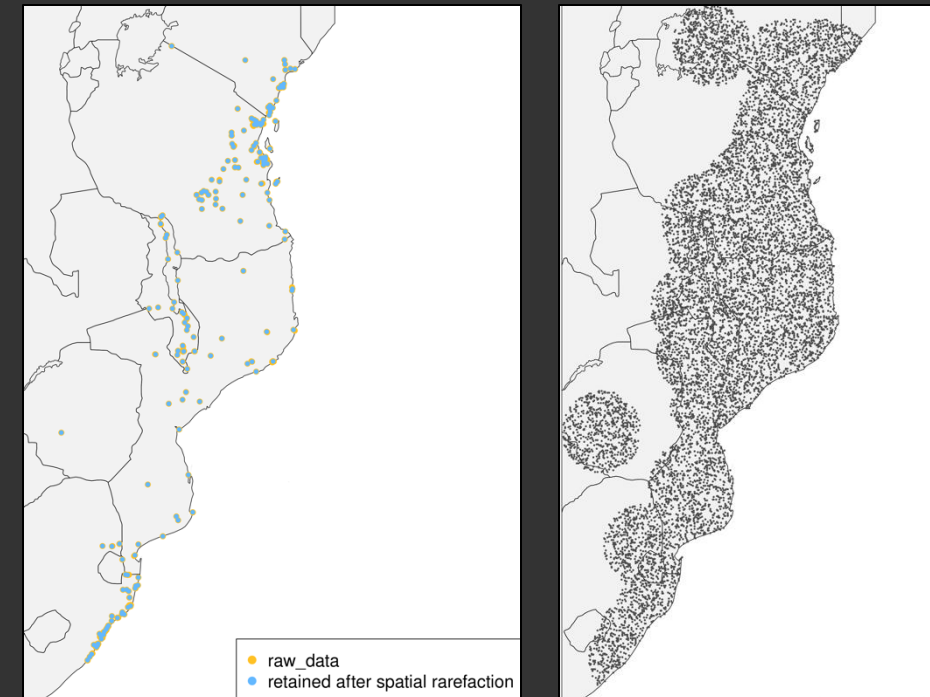
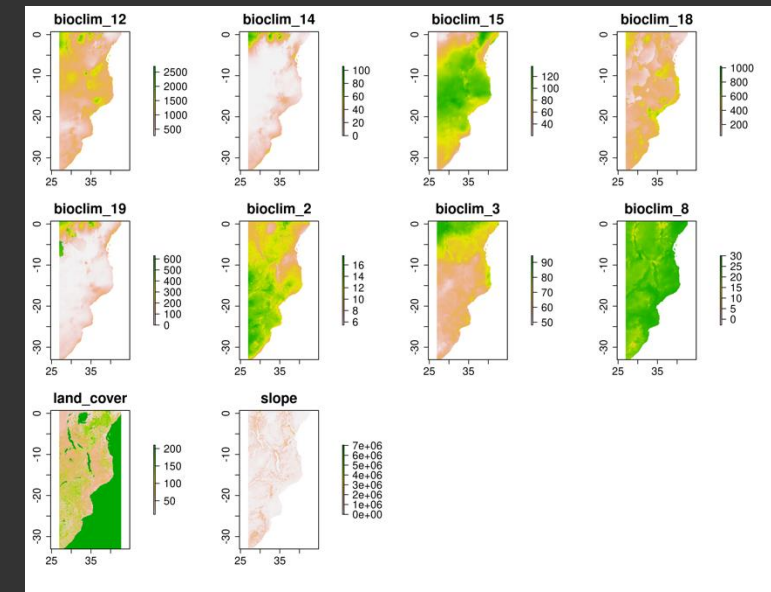
Summary.pdf



Output_log

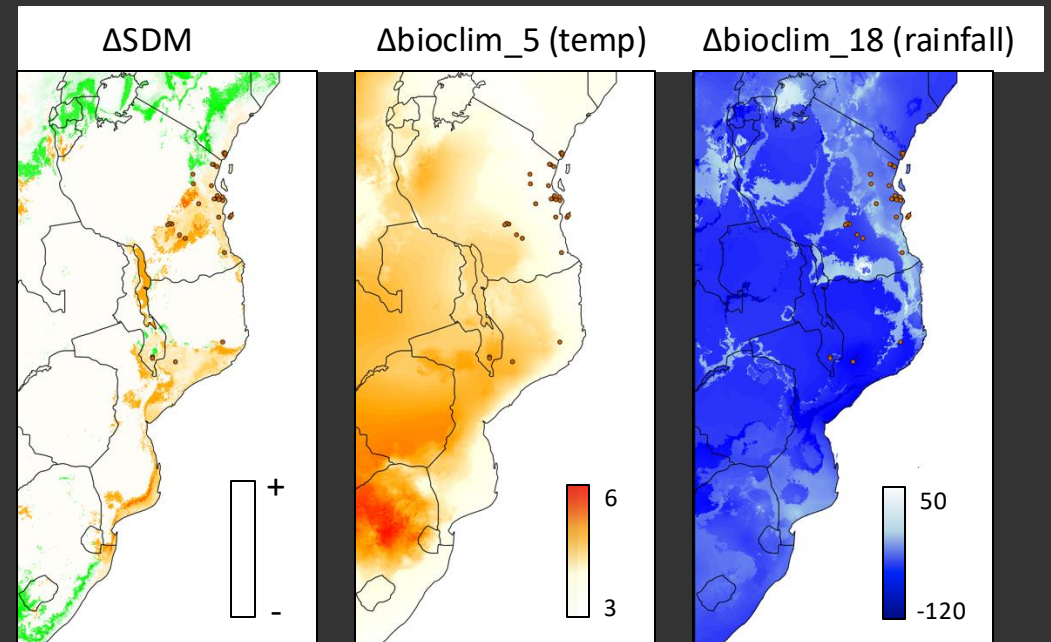
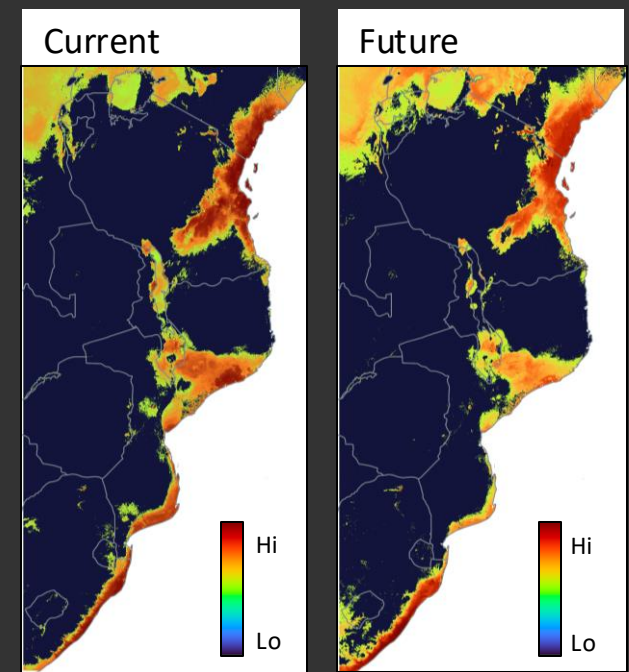
Step 0: Beating data into shape

- **Environmental data** – you supply the predictors and georeferenced data, LOtE will extract and prepare (or you can auto-download predictor data from [Worldclim2](#))
- **Species presence data** - Automated species occurrence data downloads ([rgbif](#)), cleaning ([CoordinateCleaner](#)) and spatial thinning to account for spatial autocorrelation ([spthin](#)), or you can provide your own species presence data
- **Pseudoabsence data** - Automated generation of pseudoabsence data for SDMs, or your own background data
- **All data preparation steps parameterizable** for spatial resolution, extent, defining predictors, year of prediction, spatial auto-correlation etc.



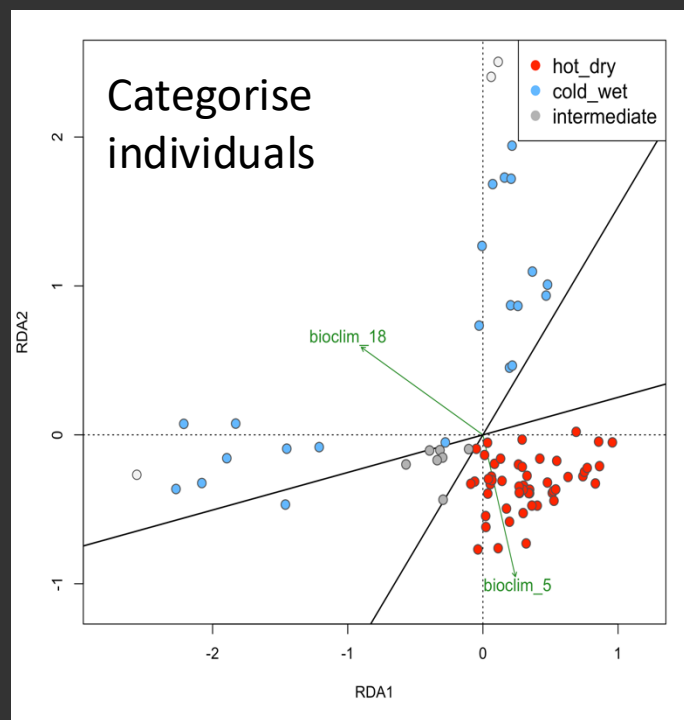
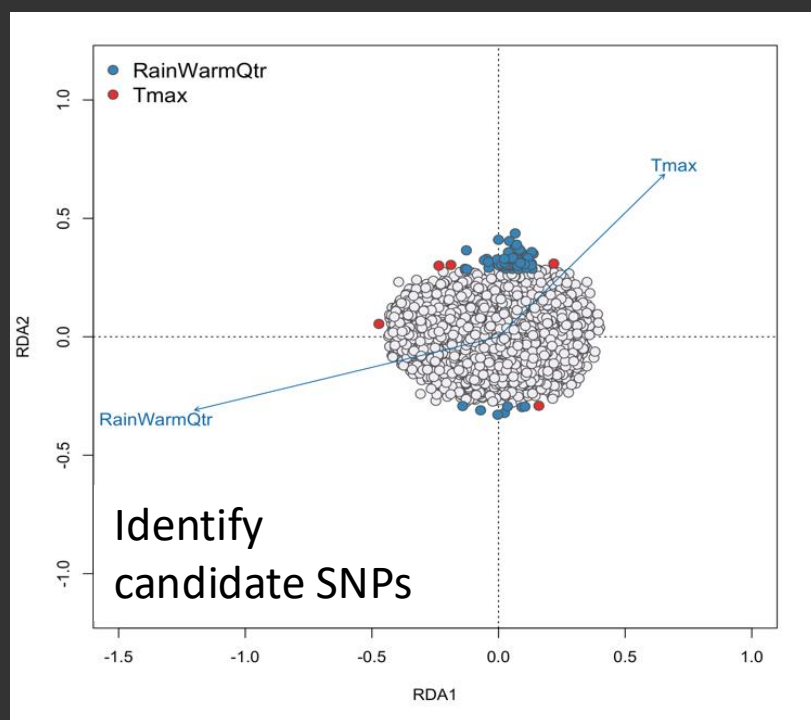
Step 1: Exposure

- **SDMs** built and evaluated (**biomod2**) based on params specified
- Ensemble SDMs projected in space and time for **current and future conditions**
- Environmental dissimilarity calculated for SDMs and environmental predictors (**raster overlay**)
- **Exposure** quantified per population based on the changes in environmental dissimilarities

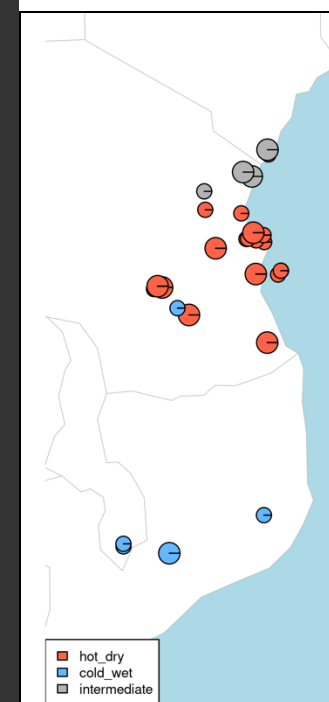


Step 2: Adaptive sensitivity

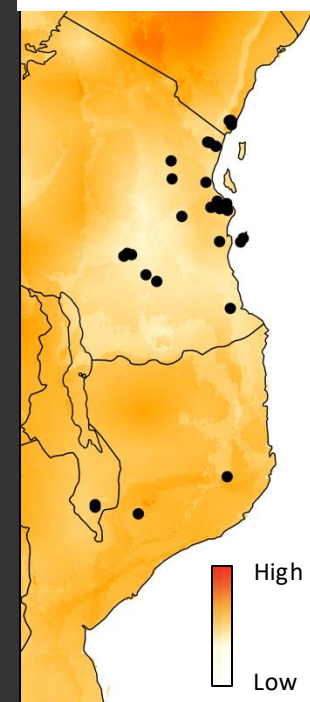
- Two GEA methods to detect candidate SNPs (RDA, LFMM). User decides thresholds (FDR, SDs from mean loadings)
- Map local adaptations in geographic space per individual/population and quantify genomic offset to quantify **adaptive sensitivity**



Map local adaptations



Quantify genomic offset



Step 3: Neutral sensitivity

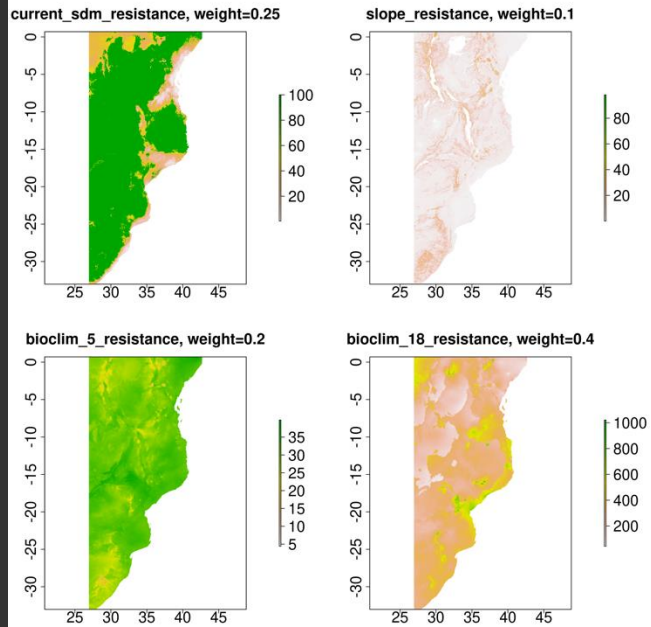
- Candidate SNPs are listed and filtered out of the genomic dataset ([Plink](#))
- **Neutral sensitivity** quantified per population based on heterozygosity across individuals ([Plink](#))

	LAT	LONG	N_individuals	Neutral_heterozygosity	Neutral_sensitivity
1	-16.000000	35.06380	13	0.43	7
2	-9.994180	39.38803	12	0.28	8
3	-7.948350	39.04373	15	0.87	4
4	-6.993650	39.29749	12	0.80	5
5	-8.348880	36.22770	10	0.71	6
6	-6.713490	38.97007	9	0.60	7
7	-7.962610	39.71429	8	0.90	5
8	-7.848980	39.79531	10	0.92	5
9	-6.136940	38.60755	12	0.67	6
10	-5.033610	38.92436	11	0.69	6
11	-6.786671	39.27574	9	0.34	8
12	-9.168889	37.03472	7	0.23	8
13	-7.179167	37.83750	6	0.44	7
14	-4.923056	38.73278	18	0.56	7
15	-4.904444	38.66250	17	0.42	7
16	-6.895222	38.81250	9	0.31	8
17	-6.901083	38.76217	12	0.41	7

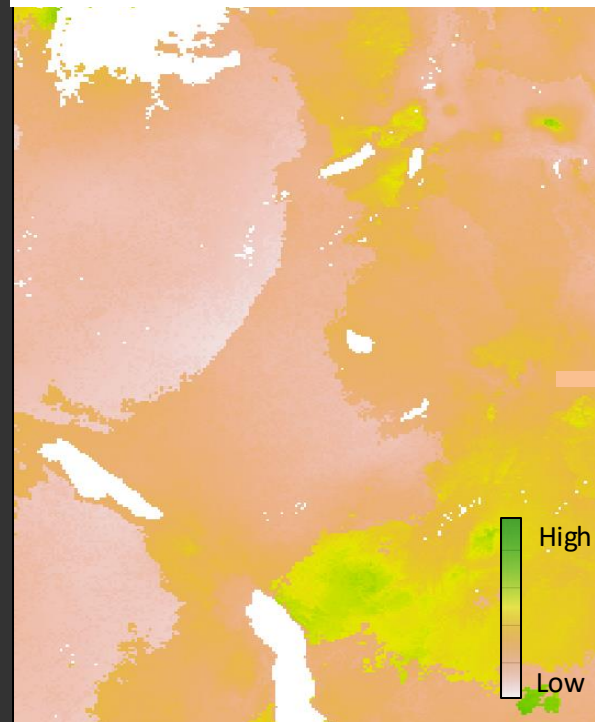
Step 4: Movement barriers

- **Population connectivity** (potential spread of individuals and genetic variants) through parameterised resistance surfaces in Julia code ([Circuitscape](#))

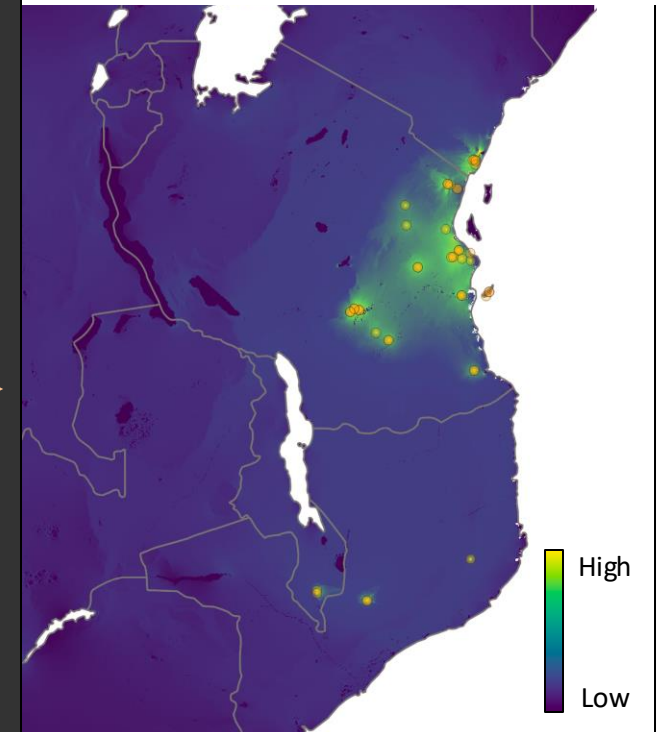
Selected variables and weights



Landscape resistance



Population connectivity



Step 5: Population vulnerability

- Integrating results from Exposure, Sensitivity and Movement barriers analyses

Exposure

LONG	LAT	env_1_current	env_1_future	env_2_current	env_2_future	sdm_current	sdm_future	exposure
35.06380	-16.00000	27.1	33.6	590	576	0.59	0.50	4.00
39.38803	-9.994180	30.7	34.4	356	632	0.55	0.00	7.00
39.04373	-7.948350	31.7	35.4	375	511	0.92	0.81	3.33
39.29749	-6.993650	31.5	35.2	280	380	0.99	0.86	3.33
36.22770	-8.348880	31.4	36.4	442	447	0.90	0.65	4.00
38.97007	-6.713490	32.0	35.6	304	409	0.97	0.81	3.33
39.71429	-7.962610	31.2	34.9	470	645	0.97	0.83	3.67
39.79531	-7.848980	31.2	34.9	475	653	0.93	0.82	3.67
38.60755	-6.136940	31.0	34.5	288	398	0.99	0.88	3.33
38.92436	-5.033610	33.1	36.7	162	225	0.96	0.88	2.67
39.27574	-6.786671	32.5	36.2	255	345	0.96	0.81	3.00
37.03472	-9.168889	30.8	35.9	467	478	0.72	0.00	6.33
37.83750	-7.179167	32.4	36.1	505	741	0.96	0.83	4.00
38.73278	-4.923056	32.1	35.7	150	211	0.98	0.85	2.67
38.66250	-4.904444	28.6	32.2	237	339	0.95	0.97	2.67
38.81250	-6.895222	31.6	35.2	296	392	0.97	0.84	3.00
38.76217	-6.901083	31.7	35.3	296	392	0.95	0.80	3.00

Sensitivity (neutral & adaptive)

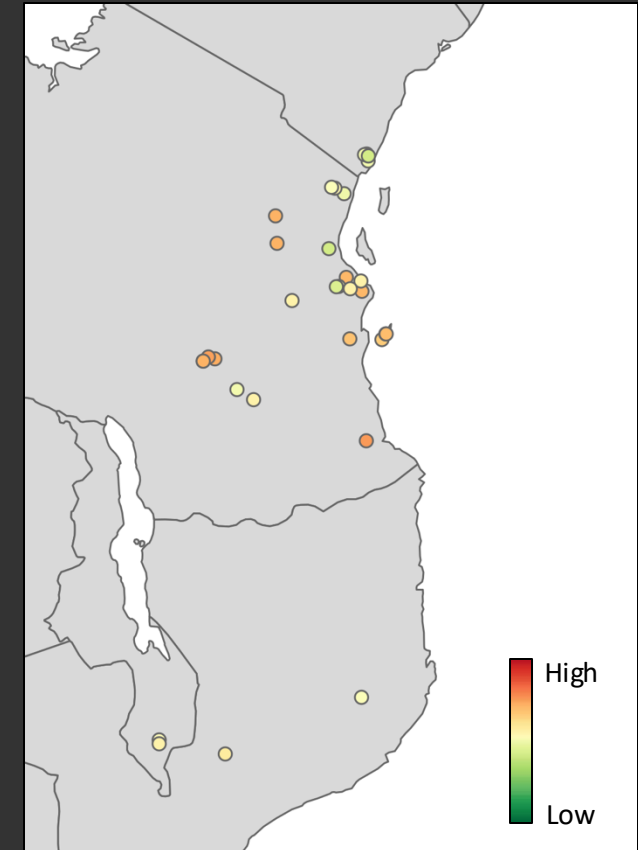
LONG	LAT	n_samples	POP_ID	hot_dry	cold_wet	intermediate	neutral_sensitivity	genomic_offset	adaptive_sensitivity
36.44378	-16.281528	2	1	0	2	0	7	2.11	10
35.06446	-16.077674	1	2	0	1	0	6	2.03	10
35.06380	-16.000000	1	3	0	1	0	7	2.09	10
39.28822	-15.144750	1	4	0	1	0	6	2.18	10
39.38803	-9.994180	2	5	2	0	0	8	1.49	8
37.03472	-9.168889	2	6	2	0	0	8	1.53	8
36.68831	-8.965944	1	7	0	1	0	6	1.48	8
35.98240	-8.395210	1	8	0	0	1	8	1.36	7
35.98376	-8.394630	1	9	0	0	1	3	1.48	8
36.22770	-8.348880	2	10	2	0	0	6	1.45	8
36.09133	-8.311798	2	11	2	0	0	7	1.50	8
39.71429	-7.962610	1	12	1	0	0	5	1.65	9
39.04373	-7.948350	2	13	2	0	0	4	1.84	10
39.80253	-7.849772	1	14	1	0	0	8	1.59	8
39.79531	-7.848980	1	15	1	0	0	5	1.61	9
37.83750	-7.179167	2	16	2	0	0	7	1.34	8
39.29749	-6.993650	1	17	1	0	0	5	1.77	9

Movement barriers

LAT	LONG	mean_connectivity	mean_connectivity_normalised	movement_barriers
36.44378	-16.281528	-0.09	0.98	1
35.06446	-16.077674	-0.26	0.80	7
35.06380	-16.000000	-0.26	0.80	7
39.28822	-15.144750	-0.75	0.27	9
39.38803	-9.994180	-0.75	0.27	9
37.03472	-9.168889	-0.75	0.27	9
36.68831	-8.965944	-0.75	0.27	9
35.98240	-8.395210	-1.00	0.00	10
35.98376	-8.394630	-0.10	0.97	4
36.22770	-8.348880	-0.10	0.97	4
36.09133	-8.311798	-0.11	0.97	5
39.71429	-7.962610	-0.10	0.98	3
39.04373	-7.948350	-0.11	0.97	5
39.80253	-7.849772	-0.10	0.97	4
39.79531	-7.848980	-0.11	0.97	5
37.83750	-7.179167	-0.10	0.97	4
39.29749	-6.993650	-0.10	0.98	2

Population vulnerability

LAT	LONG	exposure	neutral_sensitivity	adaptive_sensitivity	movement_barriers	population_vulnerability
-16.000000	35.06380	4.00	7	10	1	5.5000
-9.994180	39.38803	7.00	6	10	7	7.5000
-7.948350	39.04373	3.33	7	10	7	6.8325
-6.993650	39.29749	3.33	6	10	9	7.0825
-8.348880	36.22770	4.00	8	8	9	7.2500
-6.713490	38.97007	3.33	8	8	9	7.0825
-7.962610	39.71429	3.67	6	8	9	6.6675
-7.848980	39.79531	3.67	8	7	10	7.1675
-6.136940	38.60755	3.33	3	8	4	4.5825
-5.033610	38.92436	2.67	6	8	4	5.1675
-6.786671	39.27574	3.00	7	8	5	5.7500
-9.168889	37.03472	6.33	5	9	3	5.8325
-7.179167	37.83750	4.00	4	10	5	5.7500
-4.923056	38.73278	2.67	8	8	4	5.6675
-4.904444	38.66250	2.67	5	9	5	5.4175
-6.895222	38.81250	3.00	7	8	4	5.5000
-6.901083	38.76217	3.00	5	9	2	4.7500

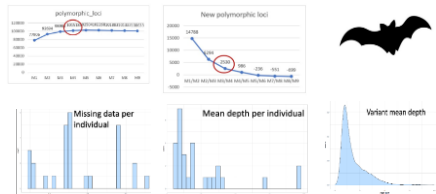


PDF Output file



Myotis_escalerae

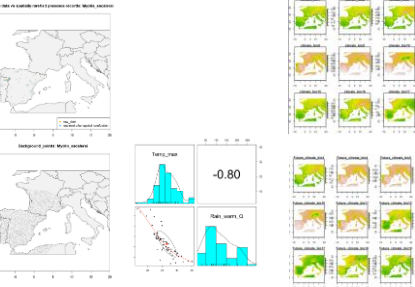
Genomic data



N genomic samples = 216
Stacks parameters used,
m= 5, M=2, n=1

Loci:
SNPs:
% missing data (inds) :
% missing data (loci) :

Spatial /environmental data



N georef samples = 216
N GBIF records = 5134
N presences = 5250

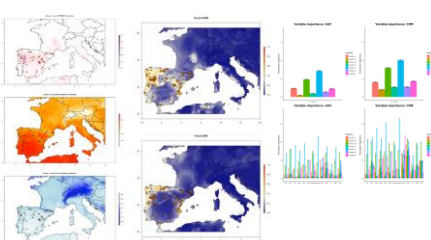
N thinned presences =
2121

N background = 2121

Current + future predictors
= blo 1,3, 4,5 6, 7 9 13, 14

Xmin=
Xmax =
Ymin =
Ymax =

Environmental dissimilarity/ SDMs



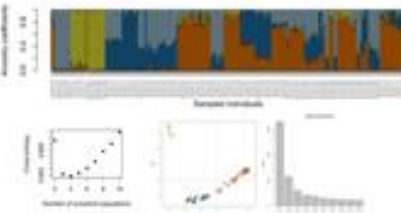
N reps = 5
Sampling = Subsample
Test/evaluation percentage
= 0.3

Evaluation:
AUC threshold = 0.8
TSS threshold = 0.5
Models passing both taken
to final ensemble = brt, rf,
maxent, gam

Variable importances:
% % % %

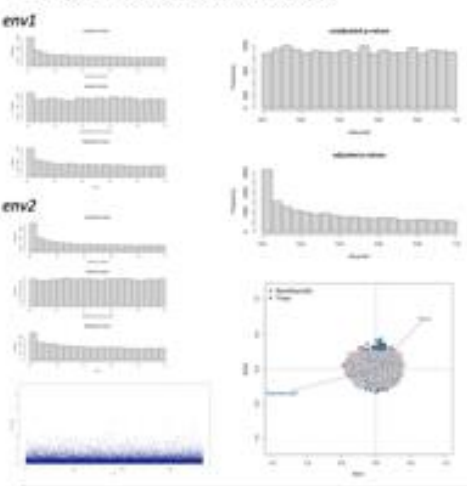
Myotis_escalerae

Population structure



N clusters sNMF = 4
N clusters PCA = 3

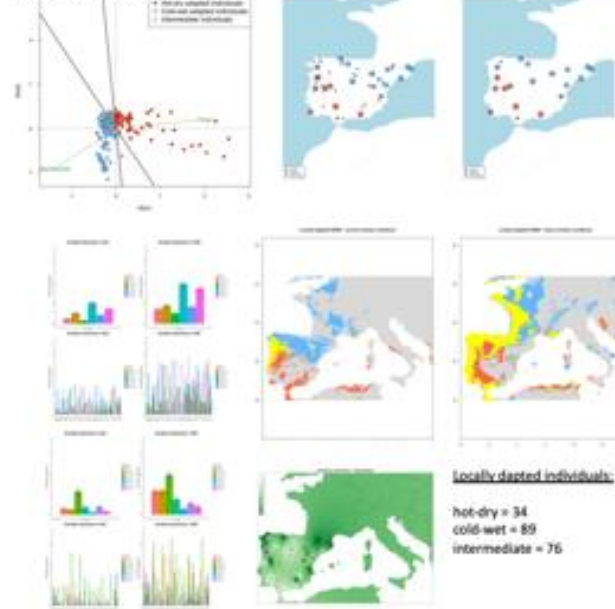
Genotype Environment Association Analysis



LFMM cand loci = 42
RDA cand loci = 27
Overlapping RDA =
LFMM = 16

Myotis_escalerae

Adaptive genetic diversity



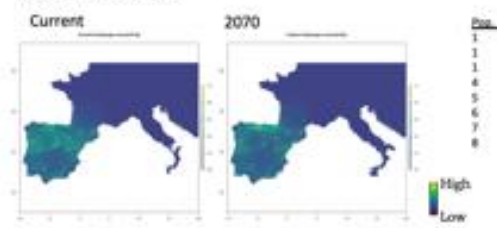
Locally adapted individuals:
hot-dry = 34
cold-wet = 89
intermediate = 76

Neutral genetic diversity



Myotis_escalerae

Range shift limitation

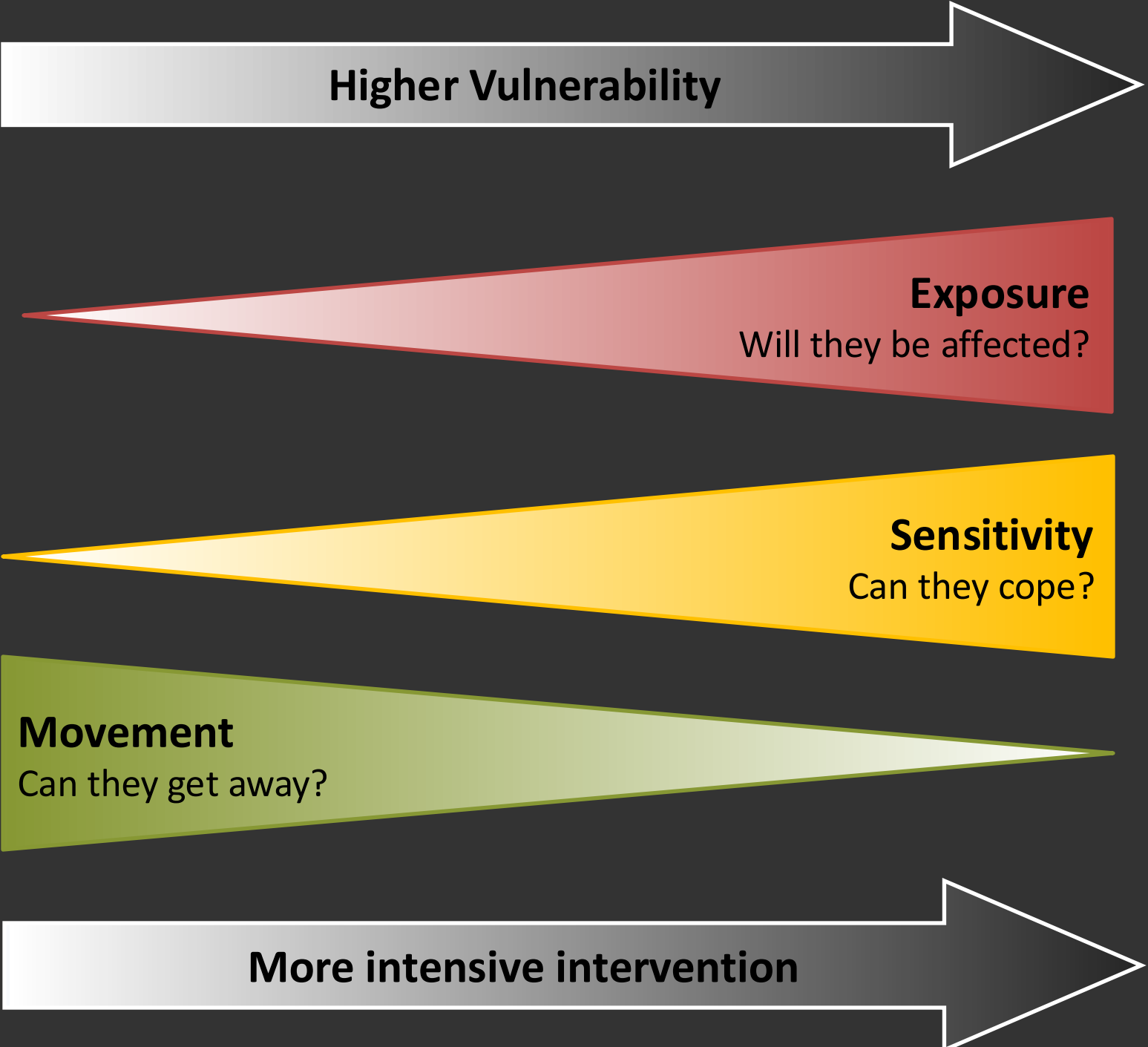


Exposure



Population vulnerability





The diagram consists of five horizontal elements stacked vertically. At the top and bottom are grey arrows pointing right, labeled 'Higher Vulnerability' and 'More intensive intervention' respectively. Between them are three colored wedges: a red wedge pointing right labeled 'Exposure', a yellow wedge pointing right labeled 'Sensitivity', and a green wedge pointing left labeled 'Movement'.

Higher Vulnerability

Exposure

Will they be affected?

Sensitivity

Can they cope?

Movement

Can they get away?

More intensive intervention

Integrating multi-species data from Europe and Africa





Chris Barratt



Methods in Ecology and Evolution

RESEARCH ARTICLE

Open Access



Life on the edge: A new toolbox for population-level climate change vulnerability assessments

Christopher D. Barratt , Renske E. Onstein, Malin L. Pinsky, Sebastian Steinfartz, Hjalmar S. Kühl, Brenna R. Forester, Orly Razgour

https://cd-barratt.github.io/Life_on_the_edge.github.io/





Tom Jenkins



Chris Barratt



Brenna Forester



Emma Stone
Tom Foxley



Kirsty Park
Michael Bekaert



Gareth Jones



<http://globalchange genetics.org>
[@orlyrazgour.bsky.social](https://www.bsky.social/orlyrazgour)



Javier Juste
Carlos Ibanez



Antton Alberdi



Stephanie Manel



Hugo Rebelo



Roberto Novella-Fernandez



Sébastien Puechmaille

Group exercise: applying LOtE

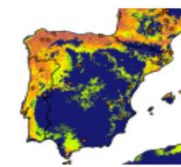
1. Case study system – species, location, geographic extent
2. Realistic sampling design – number of sites, individuals/site; how are sites/populations selected; individual vs population level study.
3. Environmental variables – SDMs, GEA / genomic offset
4. Landscape variables – movement barriers analysis.

* Justify your choices

Group discussions – 10 mins

Presentations – 1 min / group

Exposure



Predictive distribution modelling
Species distribution modelling under current and future conditions to estimate geographic range changes
[Worldclim2/CHELSA](#), [Biomod2](#)

Environmental dissimilarity
Comparison between current and future conditions to quantify magnitude of predicted environmental change

Sensitivity



Adaptive sensitivity
Identify adaptive SNPs under selection and assess spatial distribution of local adaptations and genomic offset
[LFMM](#), [RDA](#), [genomic offset](#)

Neutral sensitivity
Estimate neutral (non-adaptive) genetic diversity [Plink](#)

Movement barriers



Predictive landscape genetics
Predict population connectivity based on parameterized resistance costs of landscape
[Circuitscape](#), [Julia](#)